

Estimating in Building Construction

Chapter 14 Wood

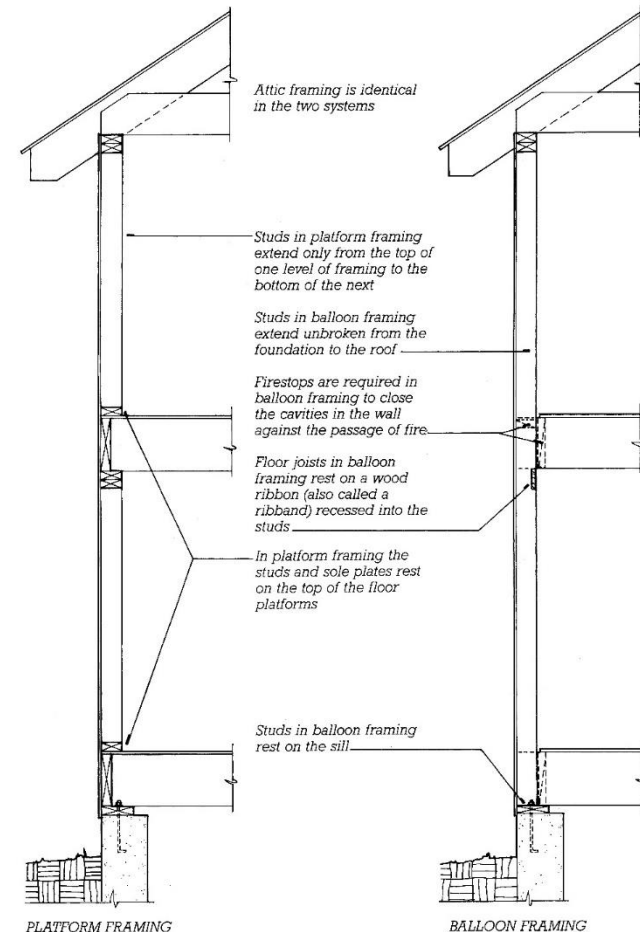
Balloon Frame

- North-American invention, 1833
- Heavy Timber: Large timbers, complicated joinery, skilled craftsman
- Light wood framing: smaller dimension wood members, simpler nailed connections, requiring less skilled builders
- Necessary technologies:
 - water- or steam-powered saw mills
 - machine-made nails
 - railroad transportation
- *Balloon Frame*: Wall studs are continuous the full height of the frame.



Platform Frame

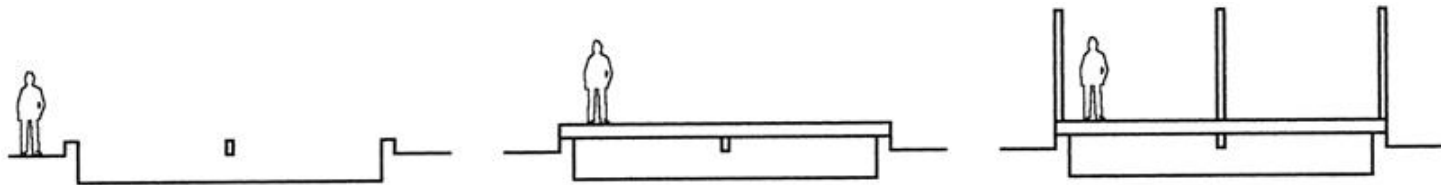
- Post-WWI variation on the balloon frame
- Relies on shorter length framing members that are easier to produce, handle, and erect.
- *Platform frame*: Single-story height wall studs span only between floor platforms



Platform Frame

Construction Concept

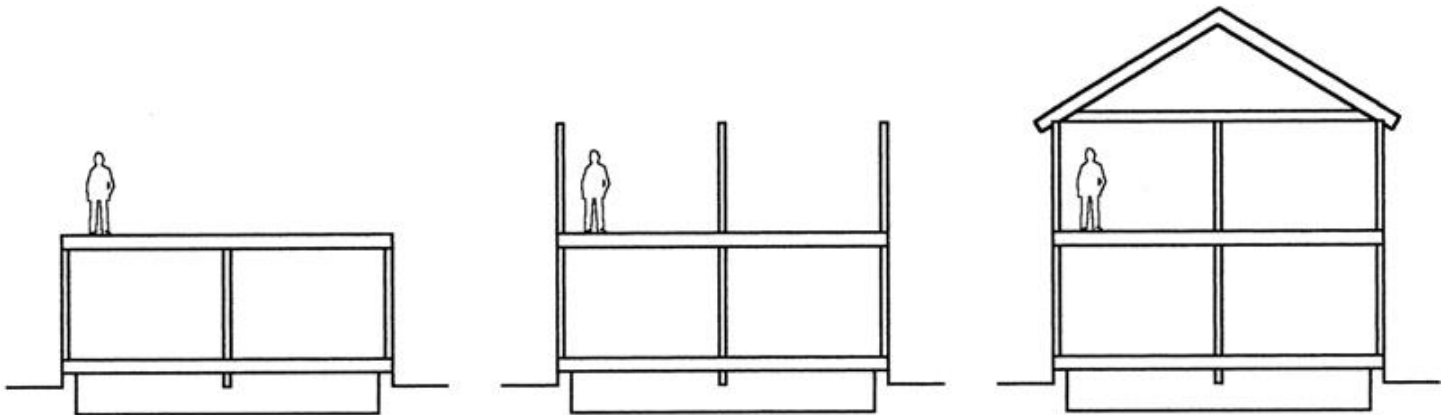
1. foundation
2. first floor platform
3. first floor walls



Platform Frame

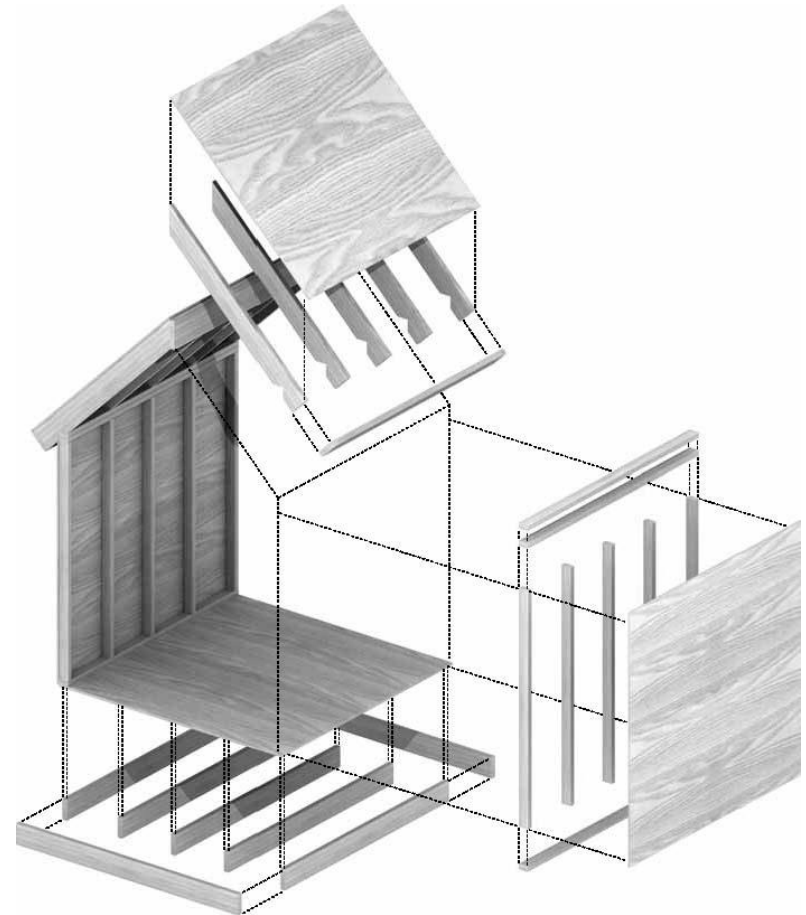
Construction Concept

1. foundation
2. first floor platform
3. first floor walls
4. second floor platform
5. second floor walls
6. roof/ceiling



Platform Frame

- Parallel nominal 2-inch members, spaced at 16" or 24":
 - Floor, ceiling *joists*
 - Wall *studs*
 - Roof *rafters*
- Plywood or OSB structural panels:
 - *Subflooring*
 - Wall and roof *sheathing*

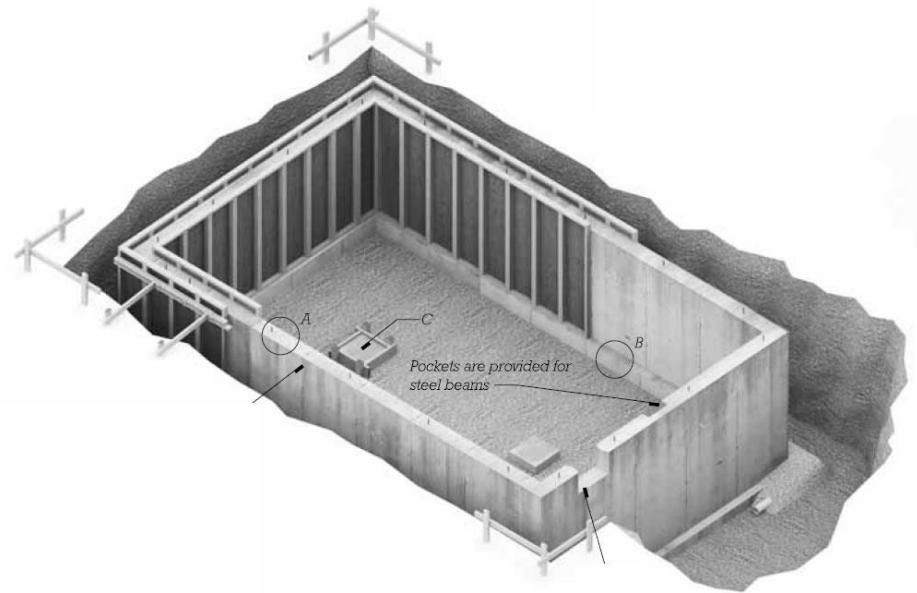




FOUNDATIONS FOR LIGHT FRAME STRUCTURES

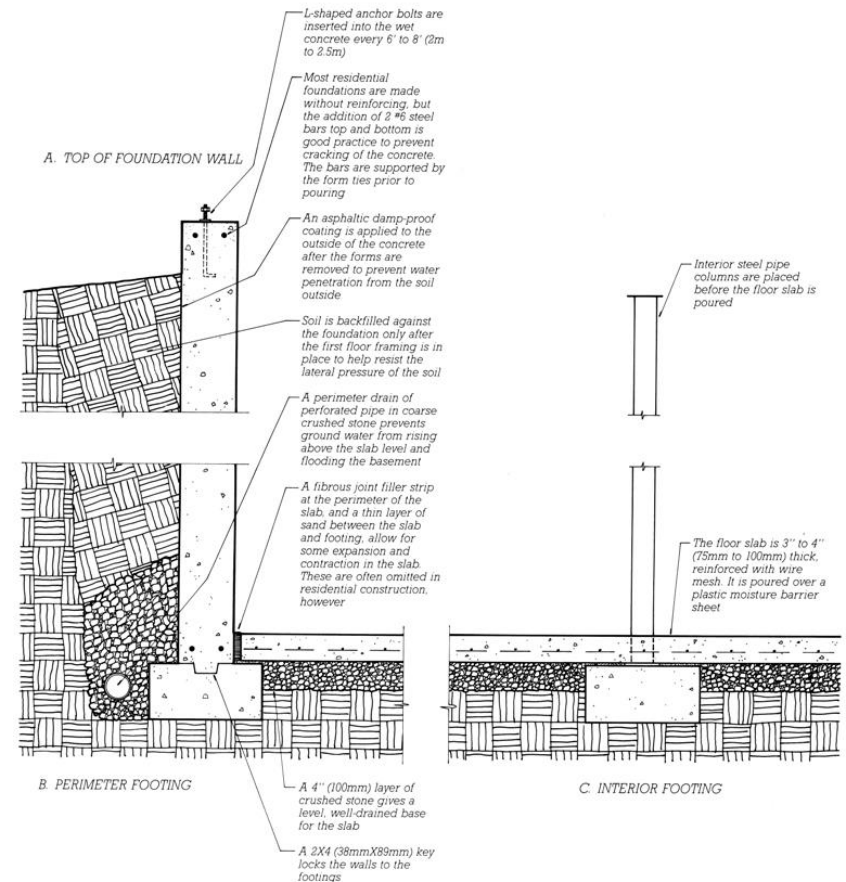
Typical Basement Foundation

- Excavation to at least below the frost line
- Cast-in-place concrete strip footing
- Cast-in-place concrete foundation wall, 8" to 12" thick, 8' to 10' tall



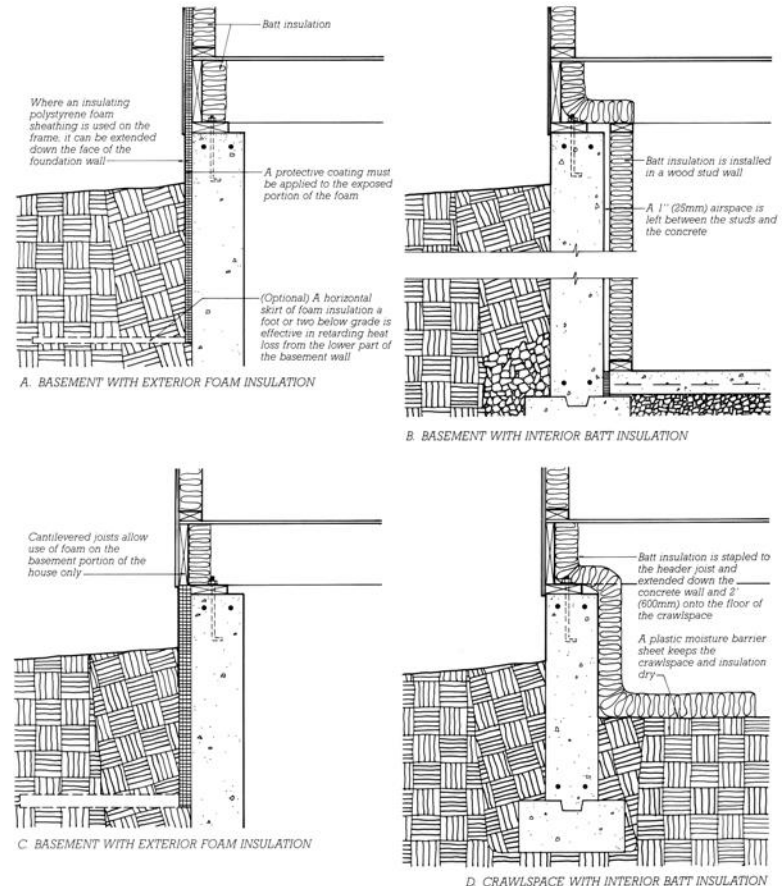
Typical Foundation

- *Anchor bolts* cast into top of foundation wall
- Steel reinforcing top and bottom of foundation wall
- Dampproofing or waterproofing on outside of foundation wall
- Crushed stone and perforated perimeter drain piping
- Backfill, after floor platform is erected to brace walls
- Slab on grade may be delayed until after structure is closed in.



Foundation Insulation

- Exterior rigid foam insulation; exposed insulation but be covered with a protective finish
- Interior glass fiber batt insulation
- Rim joist insulation

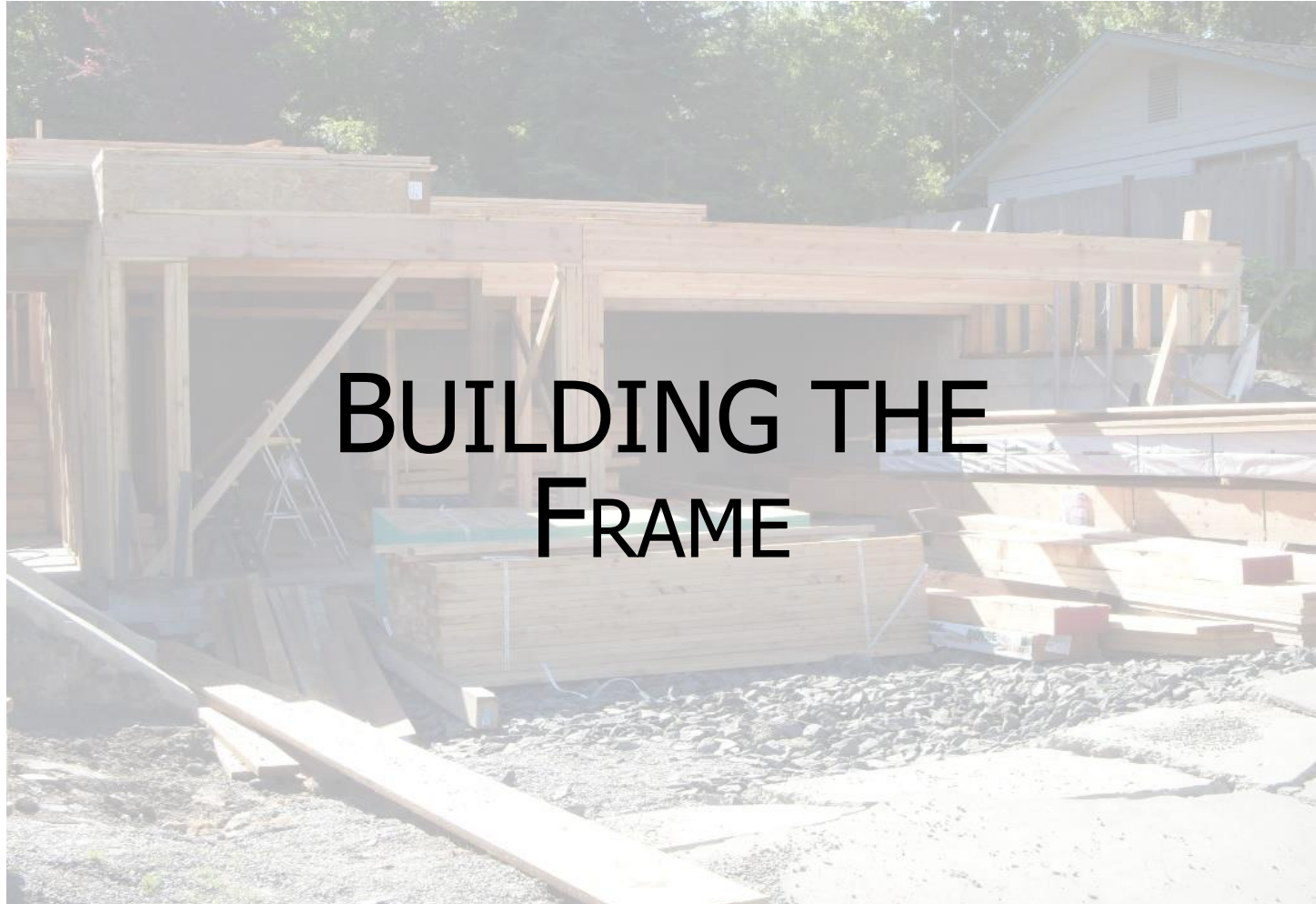


- Cast-in-place concrete foundation wall with anchor bolts protruding
- Note pockets in far side of concrete wall for setting of floor beams.



- Expanded polystyrene rigid foam foundation insulation

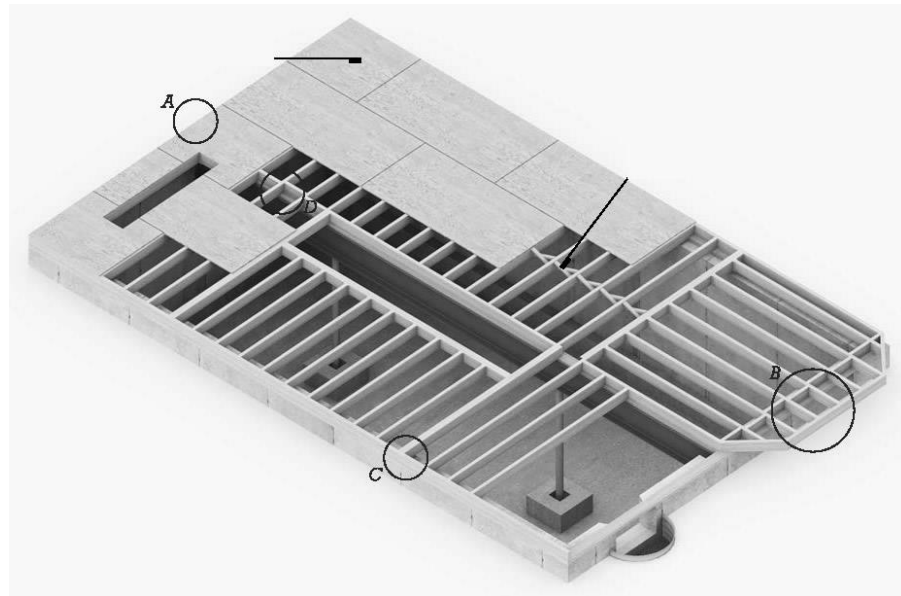




BUILDING THE FRAME

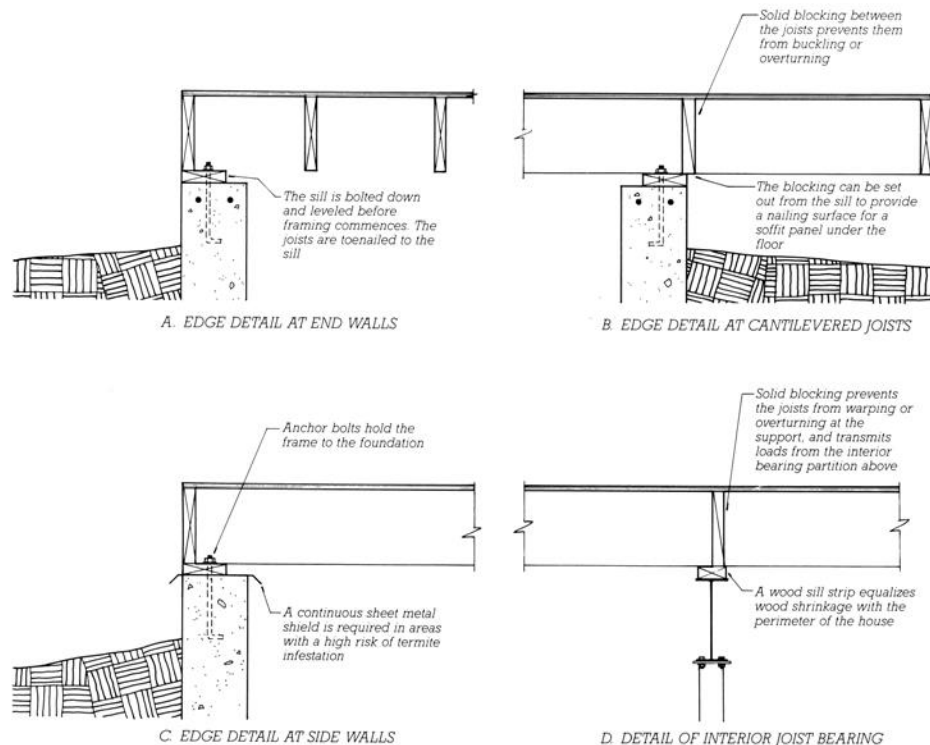
Ground Floor Platform

- *Sill plates* are anchored to foundation.
- Girders are placed.
- Floor joists are laid.
- Subflooring is applied.



Ground Floor Platform

- *Sill plates* are anchored to foundation.
- Girders are placed.
- Floor joists are laid.
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A nominal 3-inch, preservative treated sill plate is attached to the foundation with closely spaced anchor bolts. Large, square washers ensure adequate strength in the bolt-to-plate connection.



Floor framing with 2x12 solid member joists



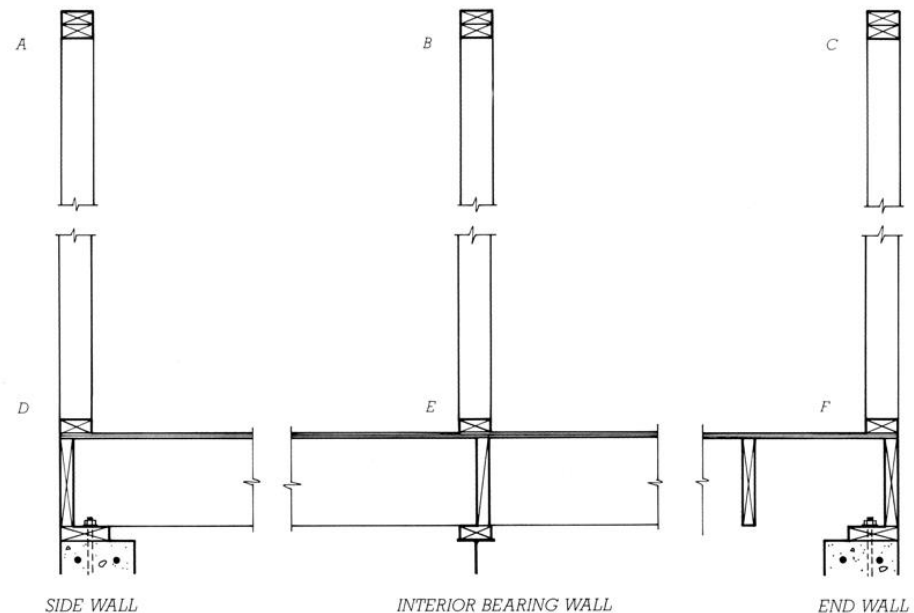
Ground Floor Walls

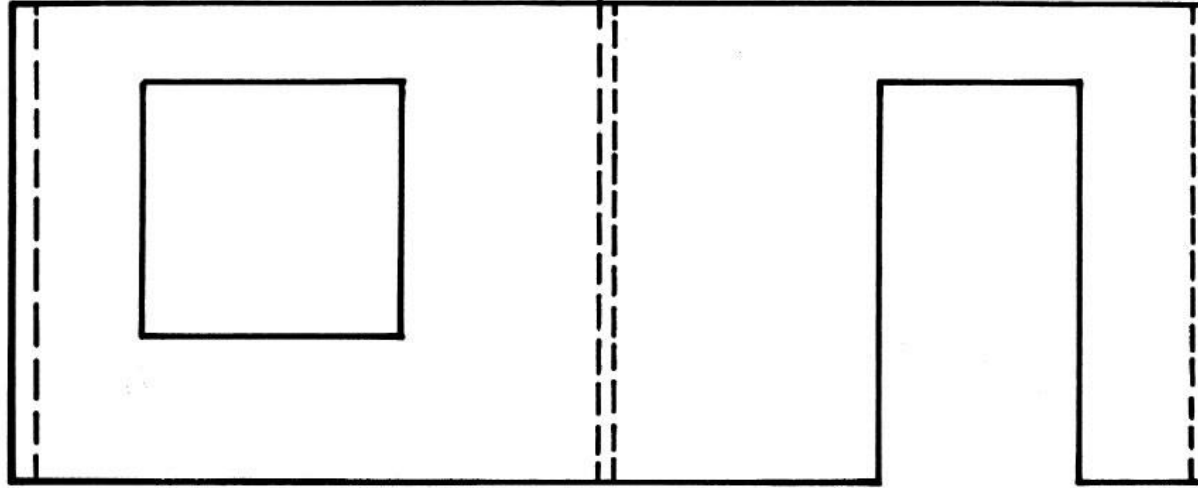
- Sections of wall are assembled lying down on the floor platform, and then are tilted into place.
- Temporary bracing holds walls in proper alignment.
- Wall sheathing may be applied before the walls are stood up, or later (as shown here), after the frame is fully erected.



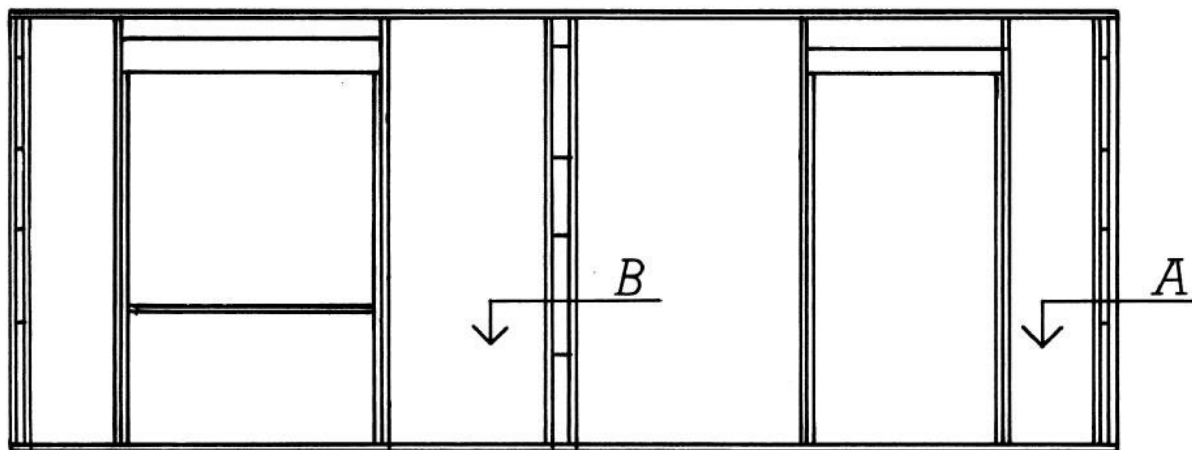
Ground Floor Walls

- *Bottom plates* are nailed to the floor platform.
- *Top plates* are doubled, to provide adequate strength to carry the joists or rafters that will be placed above. The upper top plate also overlaps and ties together adjacent sections of wall that were assembled and tilted up separately.
- Studs are continuous between top and bottom plates except at door and window openings.

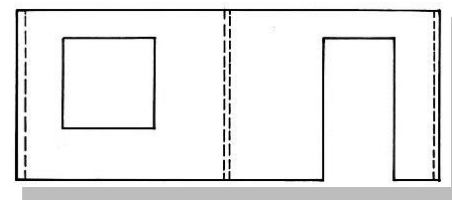




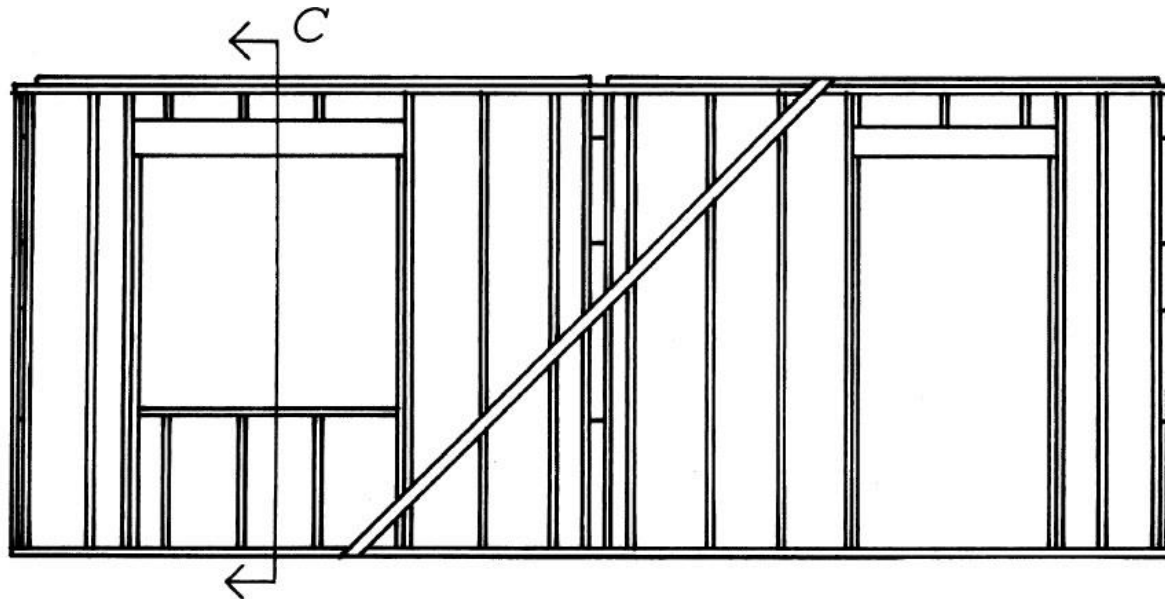
1. This is the layout of a typical exterior wall. It meets two other exterior walls at the corners, and a partition in the middle. It has two rough openings, one for a window and one for a door.



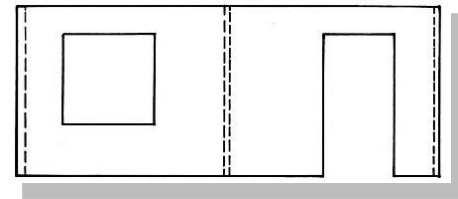
2. The framer begins by marking all the stud and opening locations on the sole plate and top plate. The "special" studs are cut and assembled first: two corner posts, a partition intersection, and full-length studs and supporting studs for the headers over the openings.

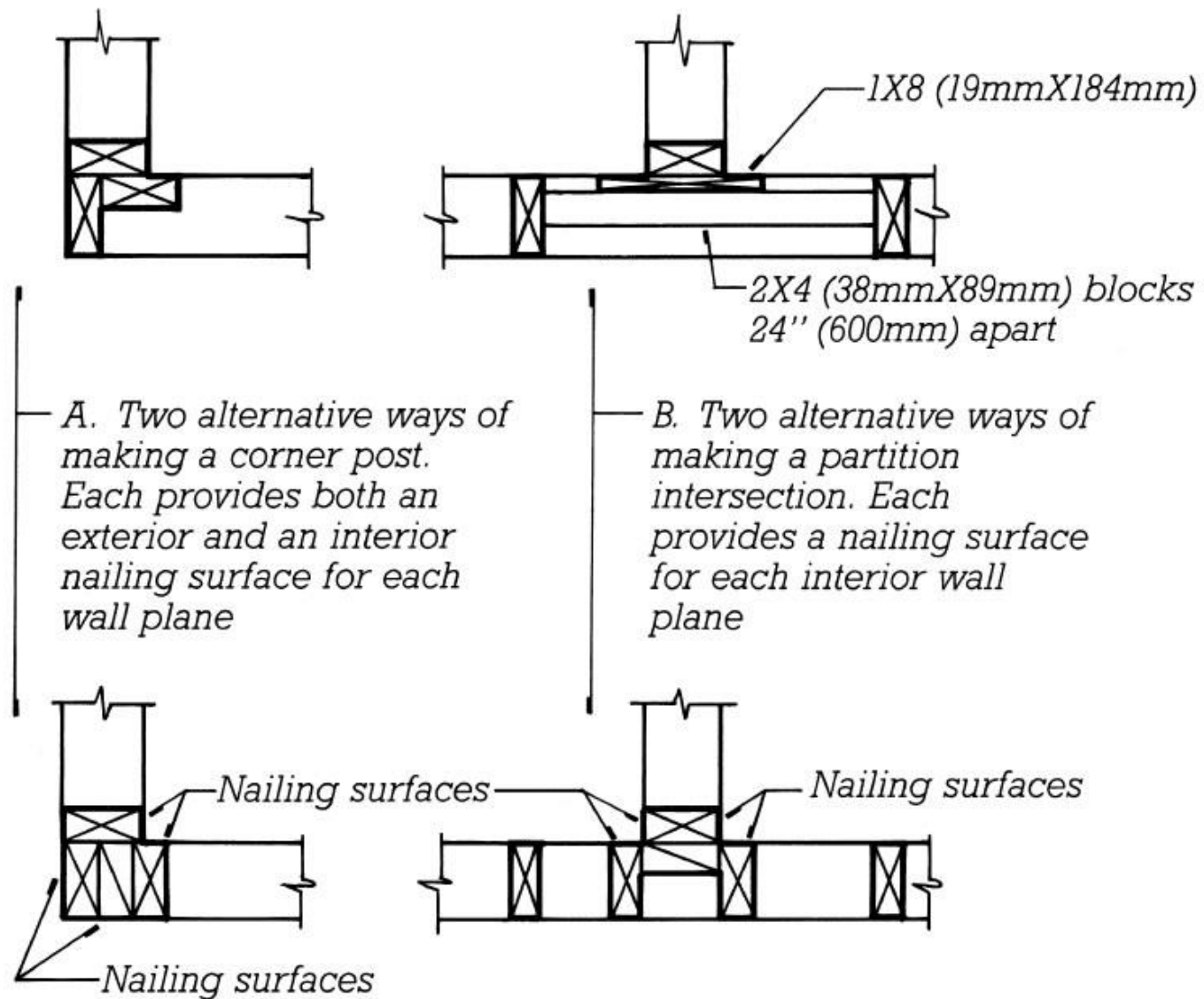


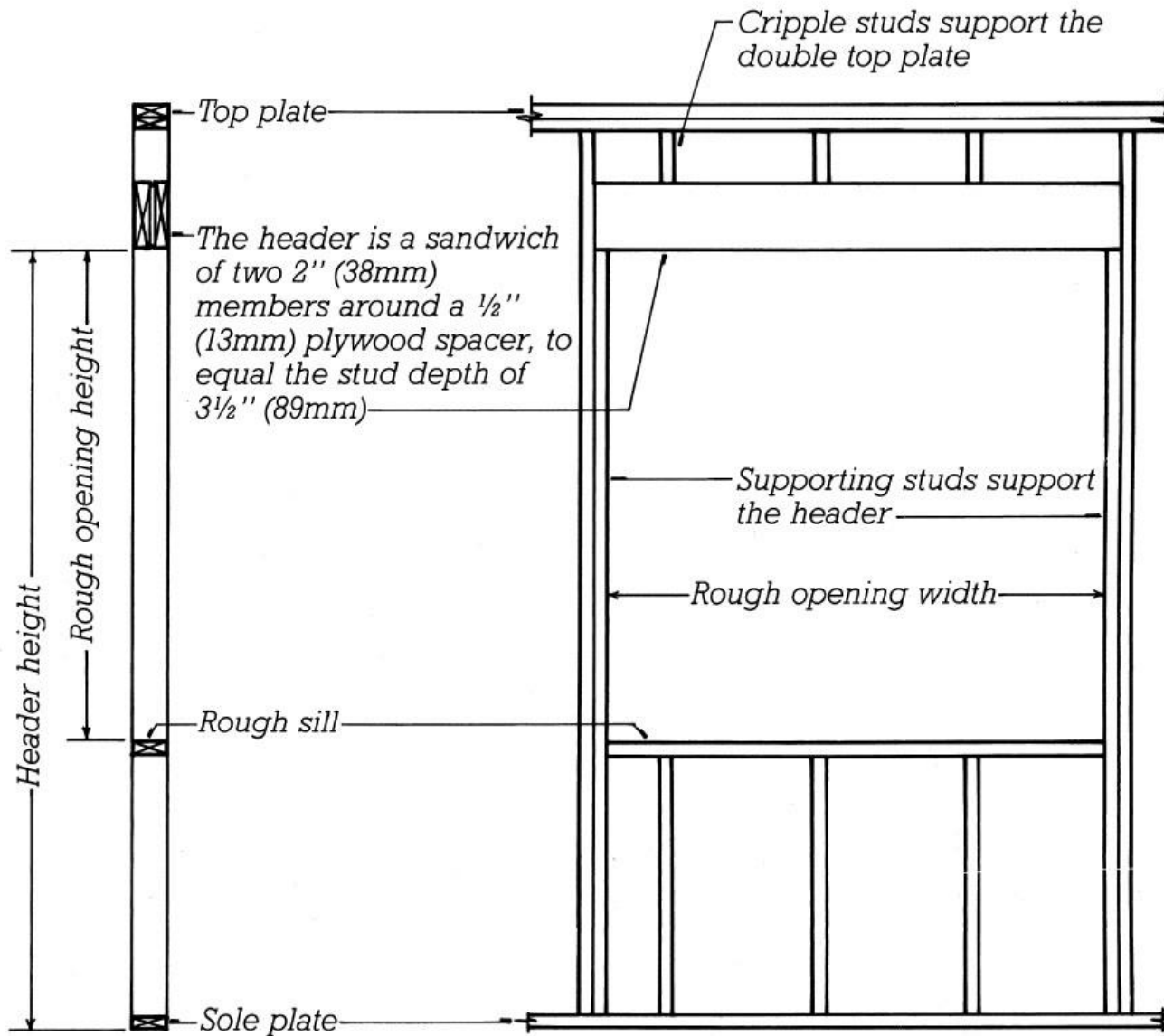




4. Diagonal bracing, usually 1X4 (19mmX89mm), is let into the face of the frame if the building will not have rigid sheathing. The second top plate may be added before the wall is tilted up, or after.







Walls are constructed in sections, lying down on the floor platform and then tilted up into position.



- Temporary bracing is used to hold the wall in place.
- Adjustable, bowed "springboards" allow more precise alignment of the walls.



Window rough opening, viewed from the inside, with the window installed:

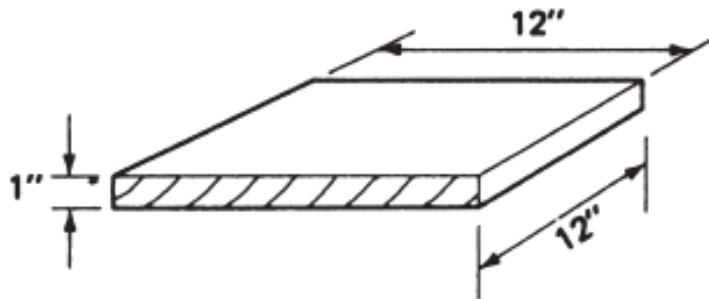
- The header is supported by two supporting "jack studs" at either end.
- The rough sill and head are also doubled.
- Note the OSB sheathing higher on the wall, and preservative-treated plywood below for greater protection from moisture damage due to differences in finish and grade conditions on the exterior side.



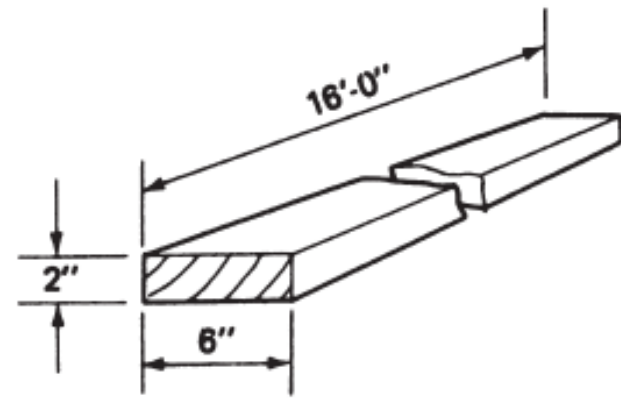
Lumber

- Bid by the piece or board measure
- Must keep sizes and lengths separate

Board Measure



A. One foot of board measure



B. Sixteen feet of board measure

Board Measure

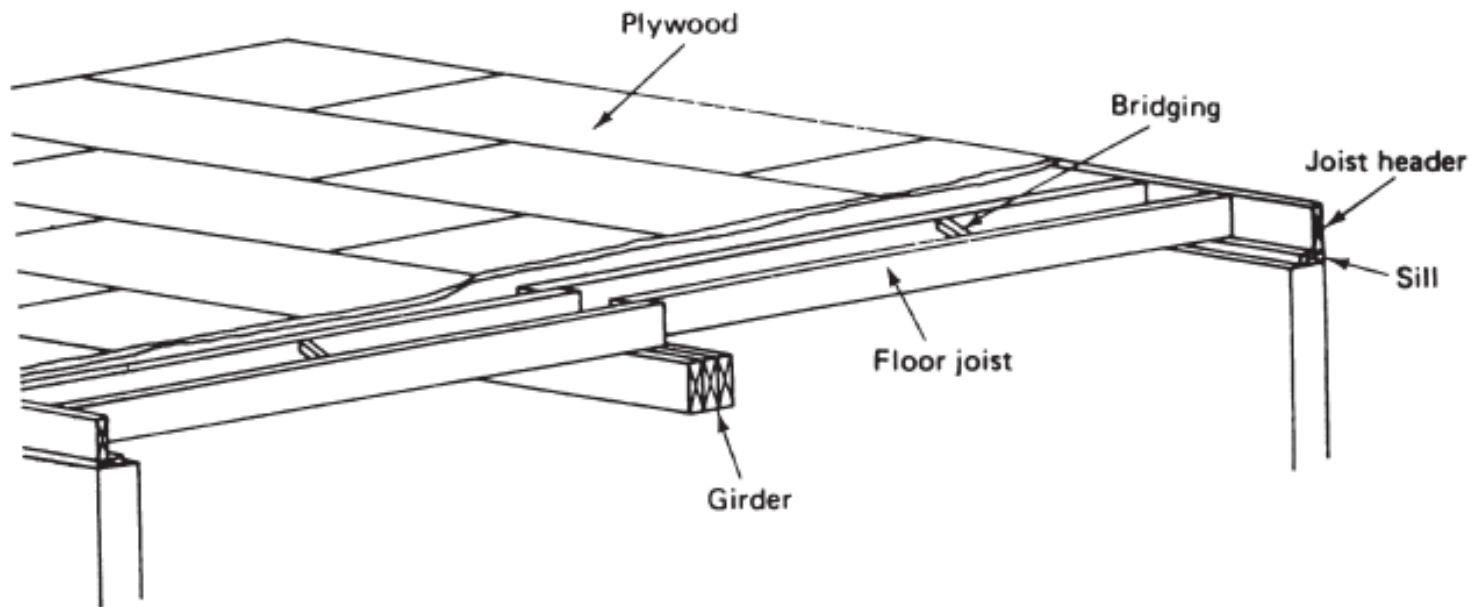
$$N = P \times T \times W \times L / 12$$

- N = Number of feet
- P = Number of pieces of lumber
- T = Thickness of lumber (in)
- W = Width of lumber (in)
- L = Length of lumber (ft)

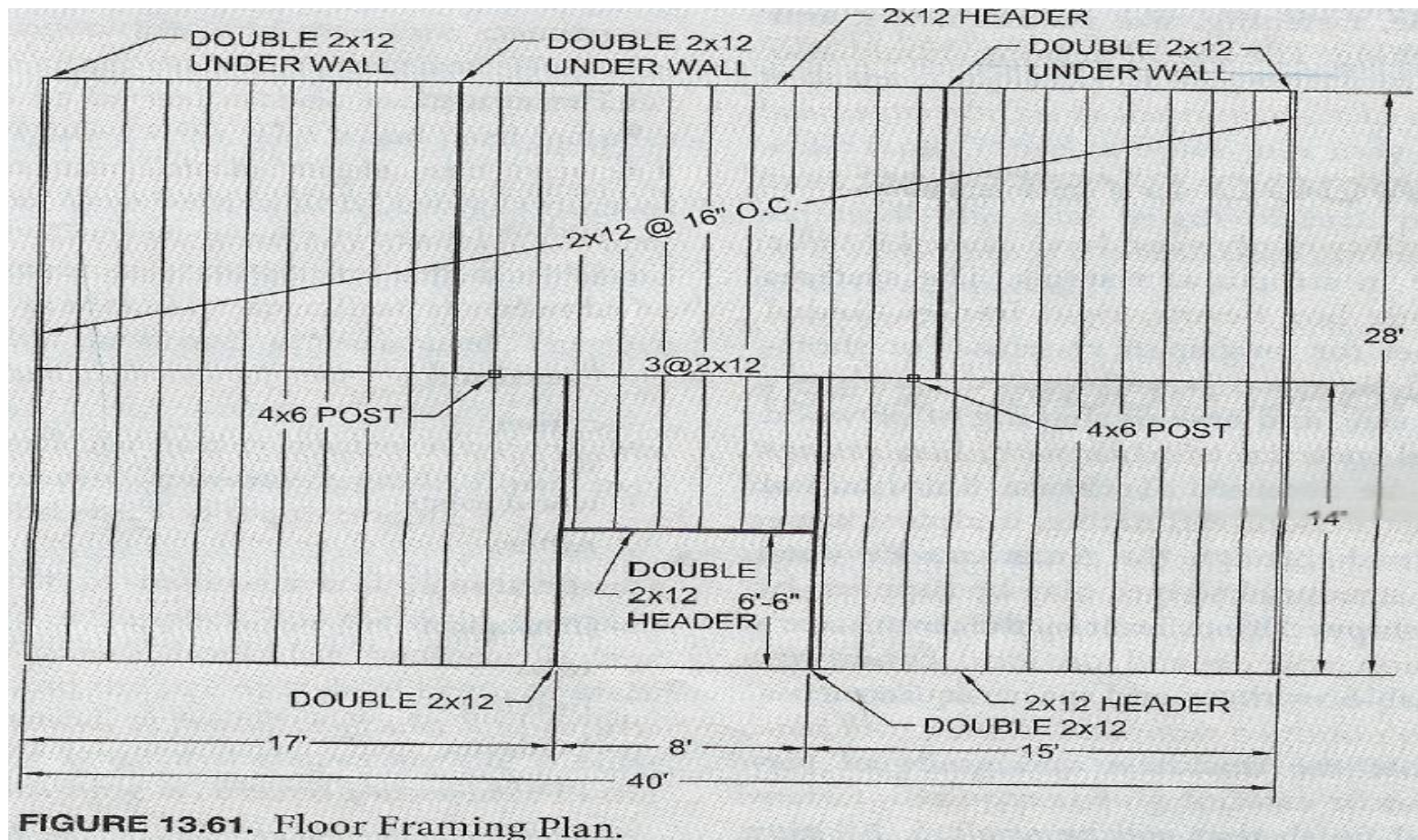
Floor Framing

- Includes:
 - Girder
 - Sill
 - Floor joist
 - Joist headers
 - Subflooring

Floor Framing



Determine the purchase quantity and number of board feet of 2×12 needed to construct the **girder** in Figure 13.61. The foundation walls are 8 inches thick and the bearing distance for the girder is 6 inches.



Determine the purchase quantity and number of board feet of 2 × 12 needed to construct the **girder** in Figure 13.61. The foundation walls are 8 inches thick and the bearing distance for the girder is 6 inches.

$$\text{Girder length (ft)} = 40' - 2'' - 2'' = 39'8''$$

$$\text{Total length of girder (ft)} = 3 \times 39'8'' = 119'$$

Order 9 pieces of 2 × 12 @ 14' so joints break over columns

$$\text{Board Feet} = 9 \times 2'' \times 12''/12 \times 14' = 252 \text{ bf}$$

Sill

- Quantity is often the same as the perimeter of the building
- Naturally durable or treated wood

A 2 × 4 **sill** plate runs around the floor in Figure 13.61. Determine the purchase quantity and number of board feet of 2 × 4 needed for the sill.

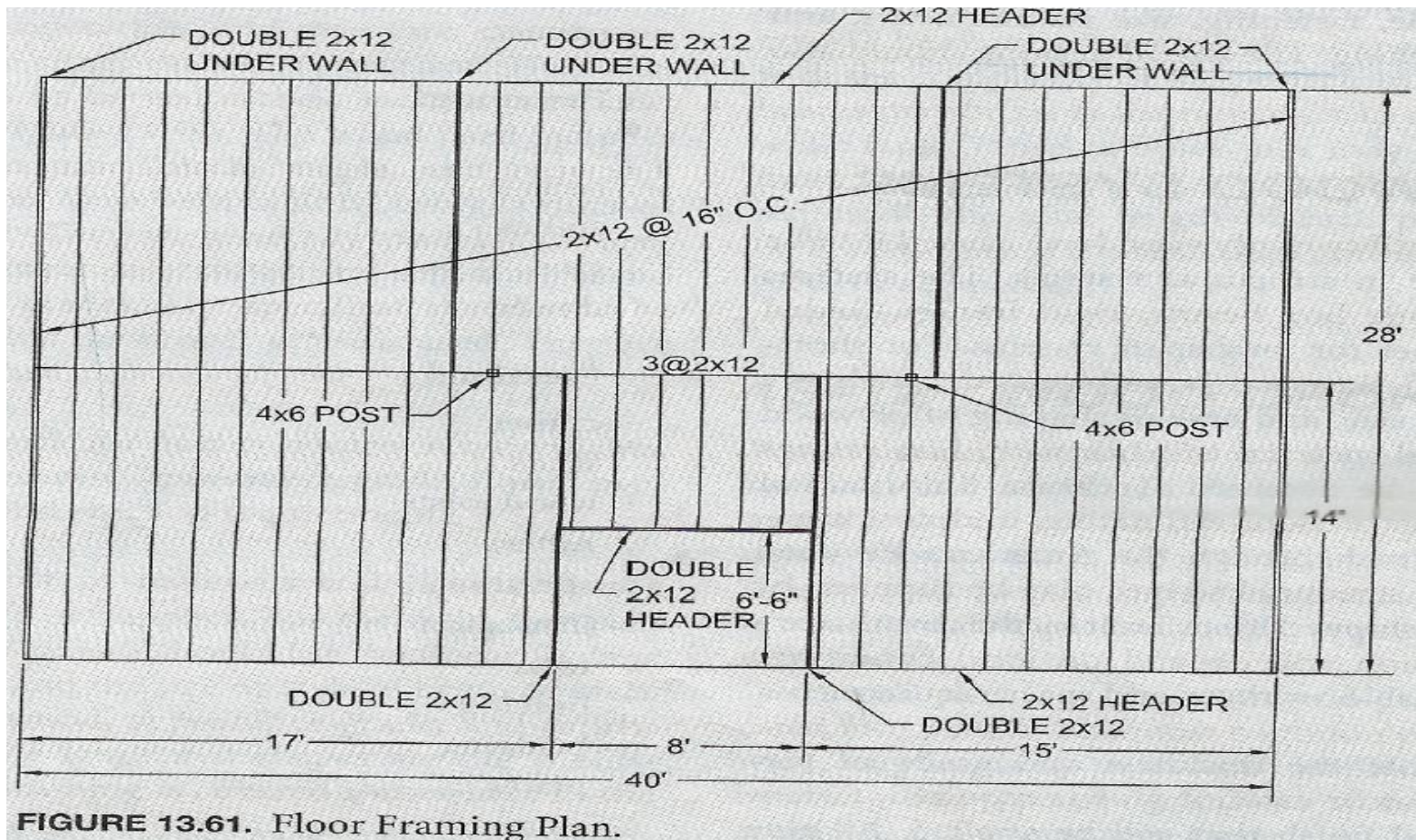


FIGURE 13.61. Floor Framing Plan.

A 2 × 4 **sill** plate runs around the floor in Figure 13.61.
Determine the purchase quantity and number of board feet of 2 × 4 needed for the sill.

Order 8 pieces 10-foot-long for the top and bottom walls and 4 pieces 14-feet-long for the side walls.

$$\text{Sill length (ft)} = 8 \times 10' + 4 \times 14' = 136'$$

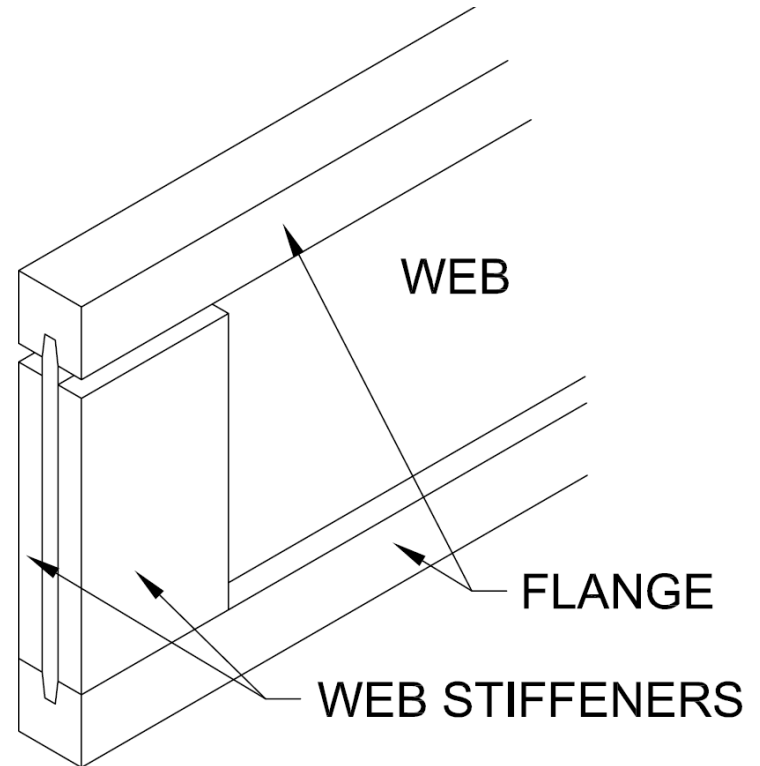
$$\text{Board Feet} = 2'' \times 4''/12 \times 136' = 91 \text{ bf}$$

Wood Floor Joists

- Lumber

- 2 x 6s, 2 x 8s,
2 x 10s, 2 x 12s

- I-joists

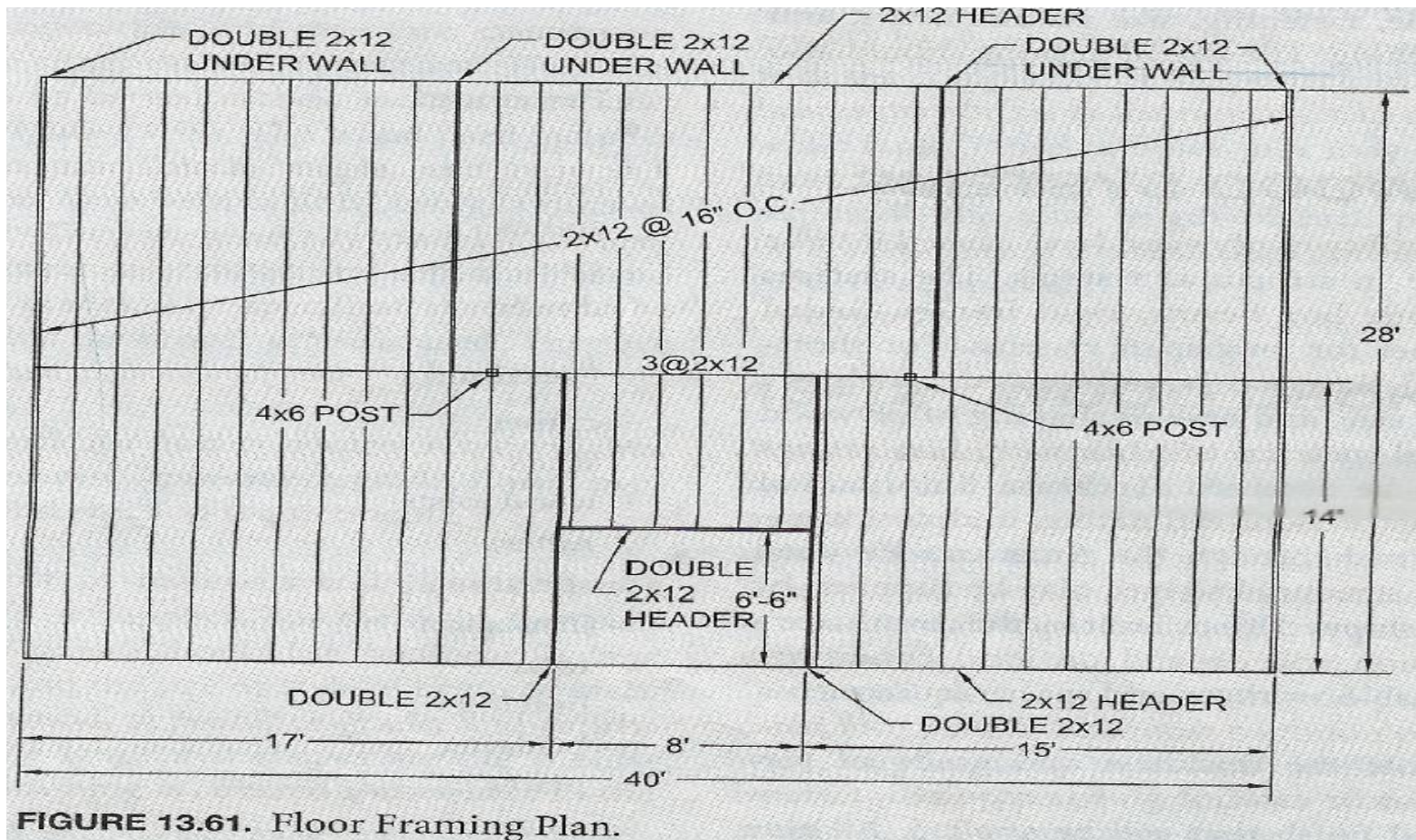


I-joist

Wood Floor Joists

O.C. Spacing (inches)	Divide length to be framed by	Multiply length to be framed by
12	1	1
16	1.33	0.75
20	1.67	0.60
24	2	0.50
30	2.5	0.40
36	3	0.33

Determine the purchase quantity and number of board feet of 2 × 12 **joist** needed for the floor in Figure 13.61. The joists are spaced 16 inches on center. Be sure to include the **headers**.



Determine the purchase quantity and number of board feet of 2 × 12 **joist** needed for the floor in Figure 13.61. The joists are spaced 16 inches on center. Be sure to include the **headers**.

Number of joist spaces = 40' × 0.75 spaces per foot = 30 spaces

Add 1 to get the number of joist – Use 31 joists.

Order 31 pieces at 14' for the top of the drawing, 25 pieces (31 – 6) at 14' for the bottom of the drawing, and 6 pieces at 8' for between the opening and the center of the building.

Order 11 additional at 14' for the double joists.

Order 8 pieces at 10' for the headers on top and bottom walls and 2 pieces at 8' for the header at the opening.

When we add all up,

67 pieces 2×12 @ 14'

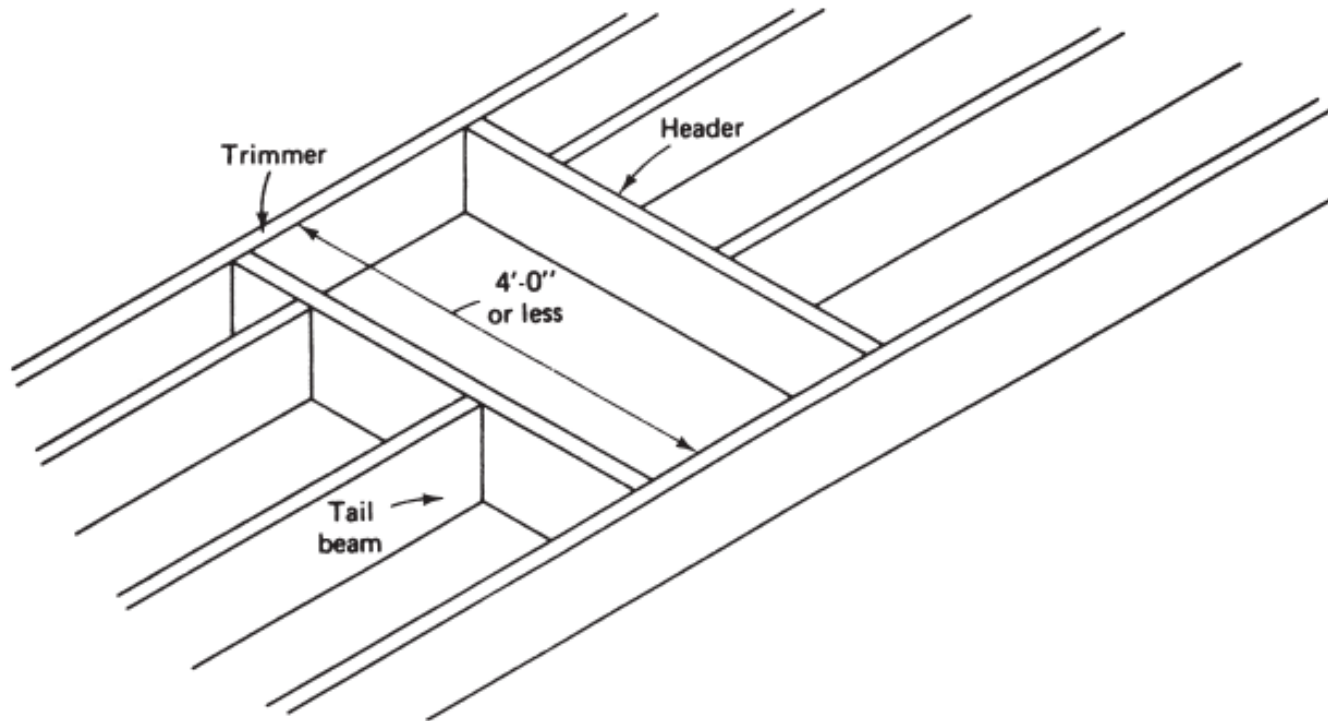
8 pieces 2×12 @ 10'

8 pieces 2×12 @ 8'

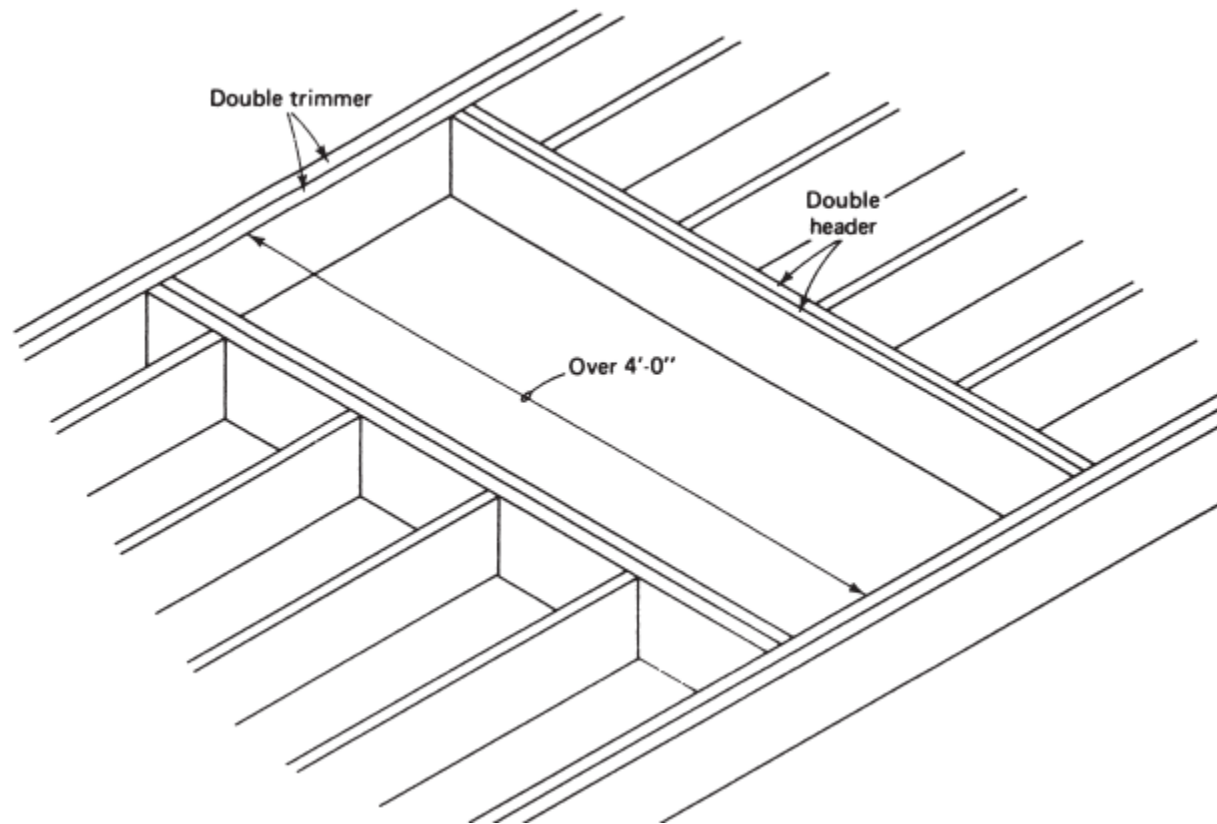
$$\begin{aligned}\text{Joist length (ft)} &= 67 \times 14' + 8 \times 10' + 8 \times 8' \\ &= 1,082'\end{aligned}$$

$$\text{Board Feet} = 2'' \times 12''/12 \times 1,082' = 2,164 \text{ bf}$$

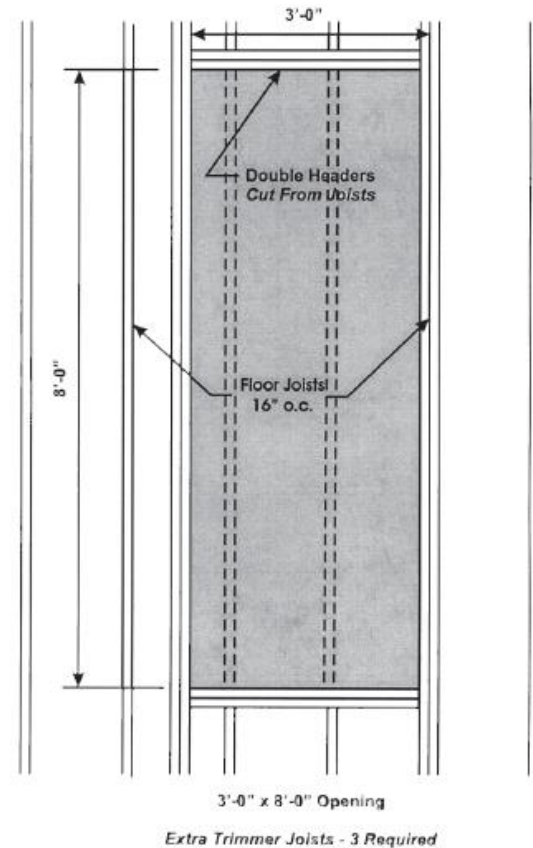
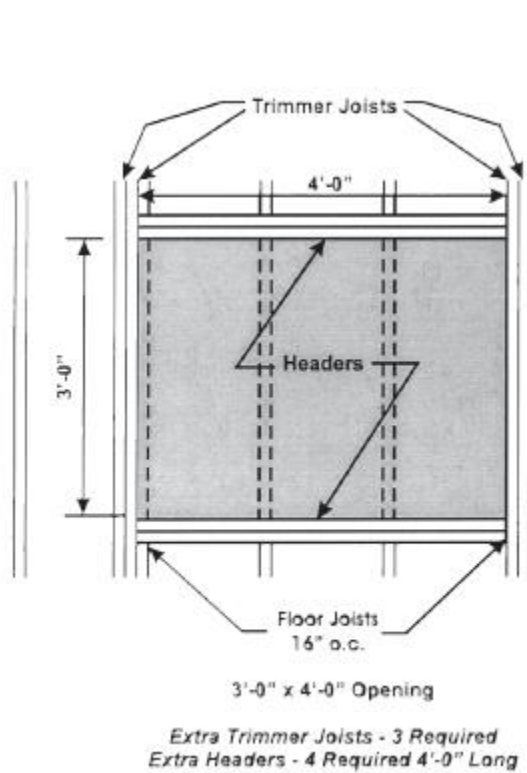
Trimmers and Headers



Trimmers and Headers

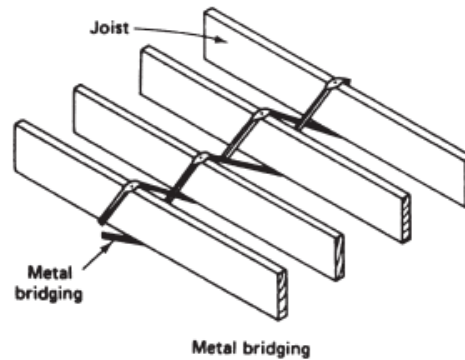
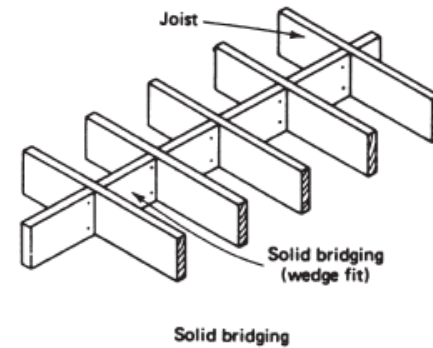
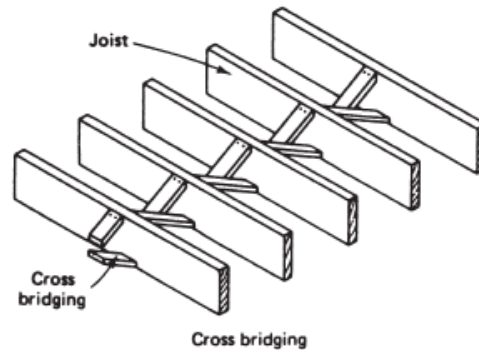


Joist Opening



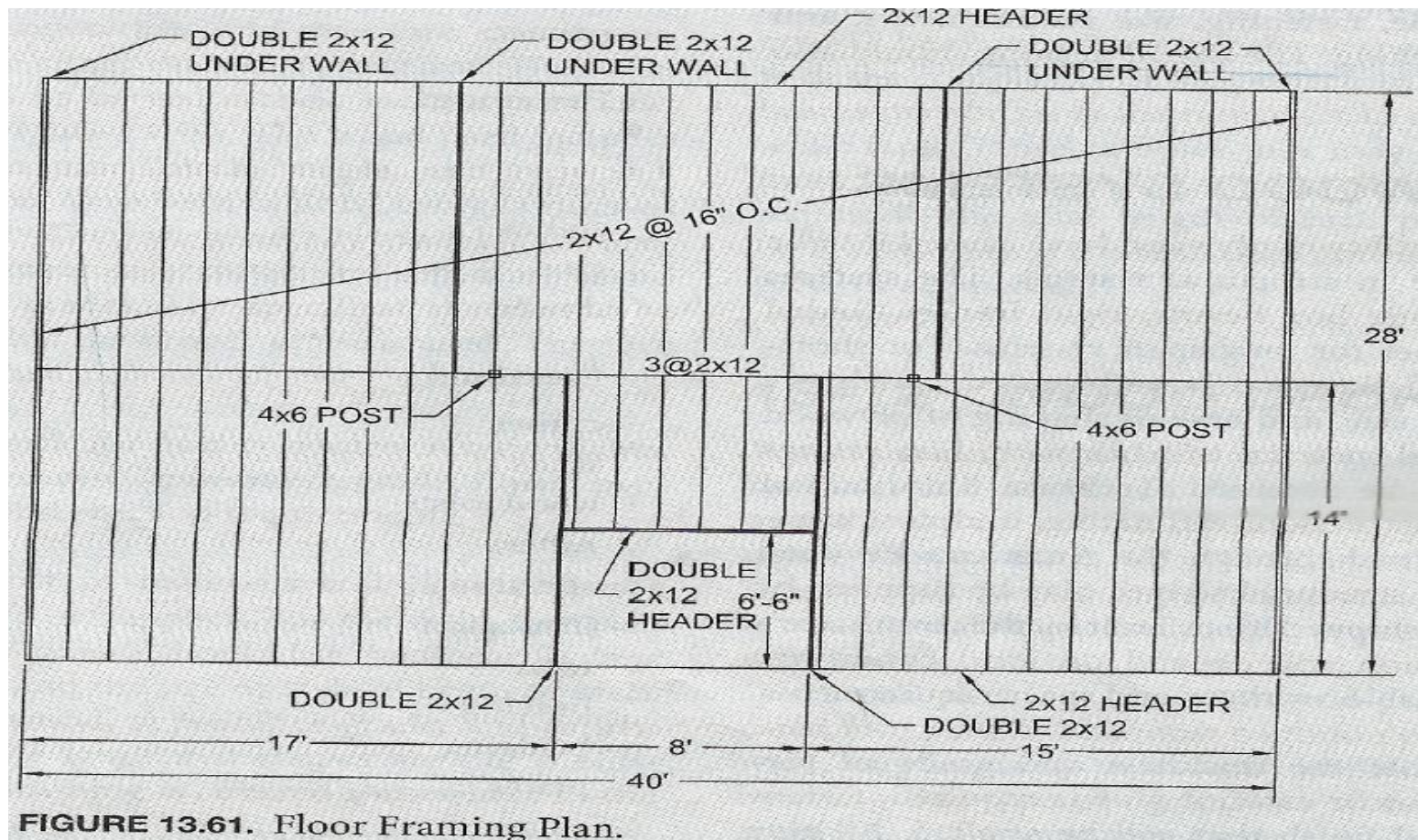
Bridging

Bridging for floor and flat roof joists and beams



Minimum sizes—wood cross bridging, 1 inch by 3 inches. Solid wood bridging, 2 inches thick, and same depth as members bridged. Metal bridging, minimum 18 gage.

Metal bridging is placed at the center of spans over 10 feet and where the joists pass over the girder. Determine the number of pieces of **metal bridging** needed for the floor in Figure 13.61.



Metal bridging is placed at the center of spans over 10 feet and where the joists pass over the girder. Determine the number of pieces of **metal bridging** needed for the floor in Figure 13.61.

30 pairs in the top half of the drawing

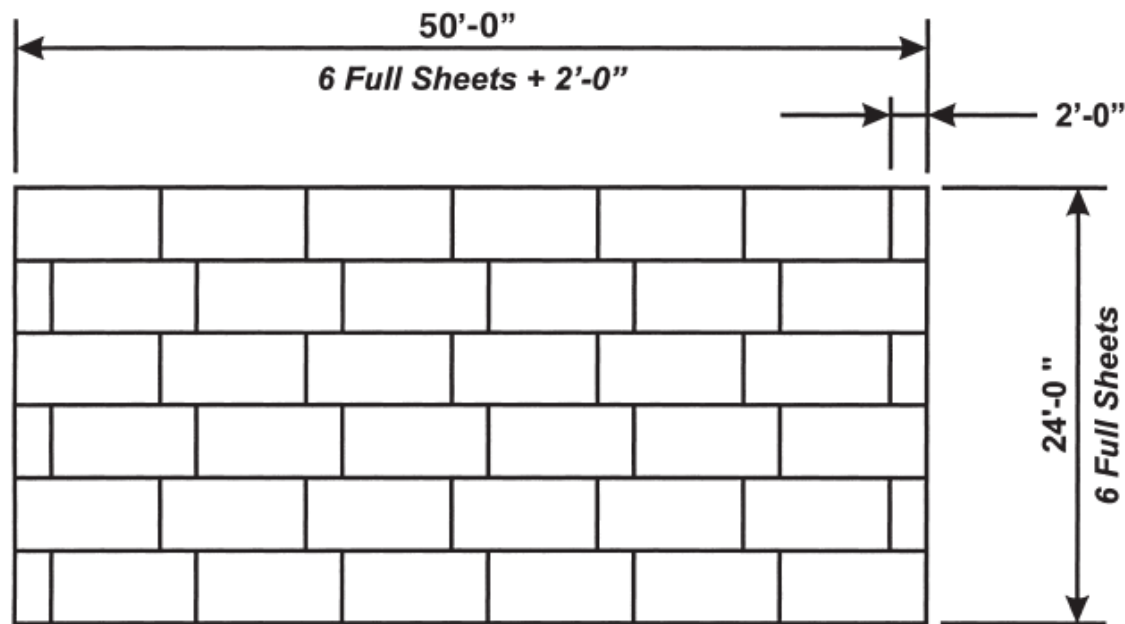
30 pairs over the girder

24 pairs in the bottom half of the drawing

Number = $2 \times (30 + 30 + 24) = 168$ each

Subflooring

- Bid by the sheet



**36 Full Sheets Plus 6 - 2'-0" Pieces = 37 ½ Sheets
Order 38 Sheets**

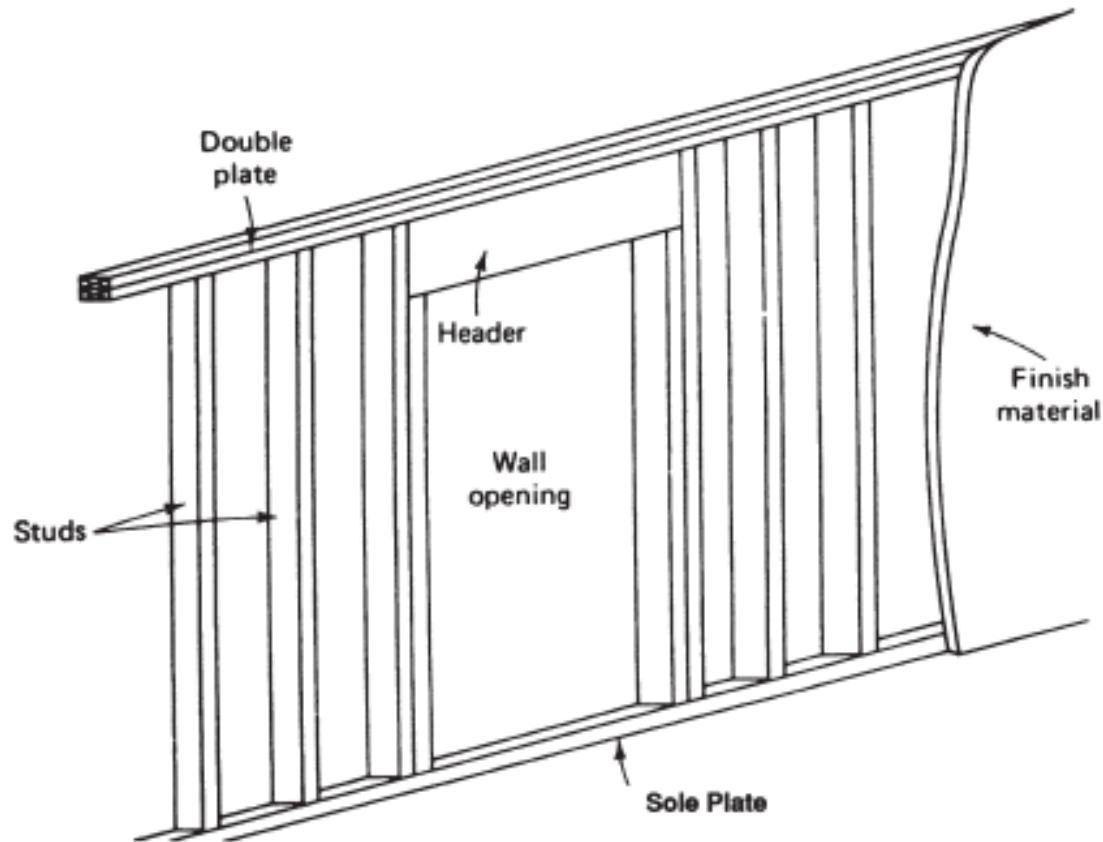
Determine the number of 4-foot by 8-foot **sheets of plywood** needed for the floor in Figure 13.61. Ignore the opening.

$$\begin{aligned}\text{Sheets of plywood} &= 40' \times 28' / 32 \text{ sf/sheet} \\ &= 35 \text{ sheets}\end{aligned}$$

Wall Framing

- Includes
 - Sole plates
 - Studs
 - Top plates (double?)
 - Headers
 - Finish materials

Wall Framing



Bottom Plates

- Bid by the
 - Piece
 - Lineal foot
 - Board measure

Plates

- Bottom plates:
 - Equals the length of the wall
 - Must be of naturally durable or treated wood when on concrete or masonry

- Double top plates
 - Equals twice the length of the wall

Determine the purchase quantity of 2 × 4 **top and bottom plates** needed for the **exterior wall**.

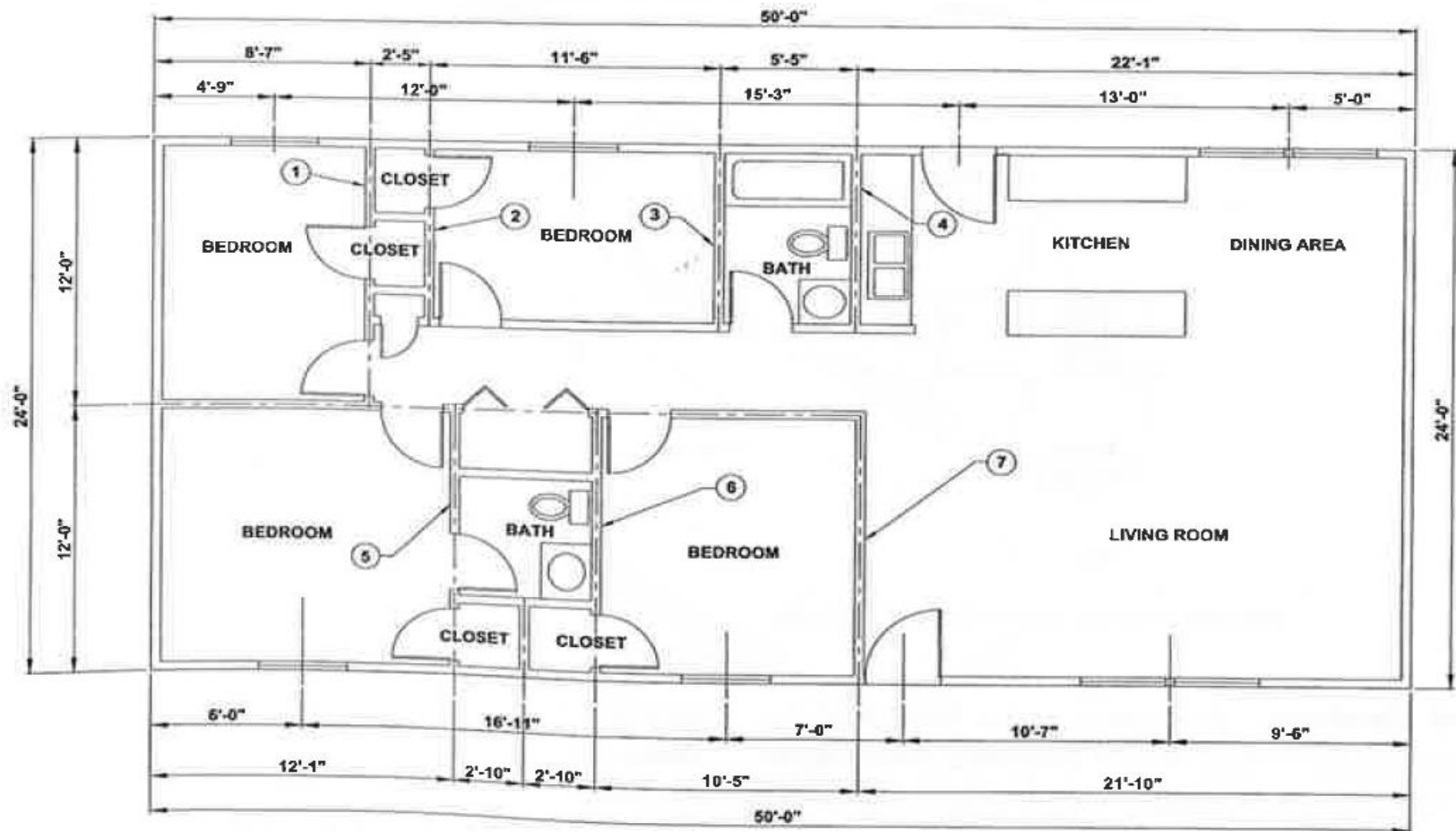


FIGURE 13.9. Floor Plan.

Purchase Quantity

10 pieces 2 × 4 treated @ 10'

4 pieces 2 × 4 treated @ 12'

20 pieces 2 × 4 @ 10'

8 pieces 2 × 4 @ 12'

Studs

- Must be keep separate by size and length

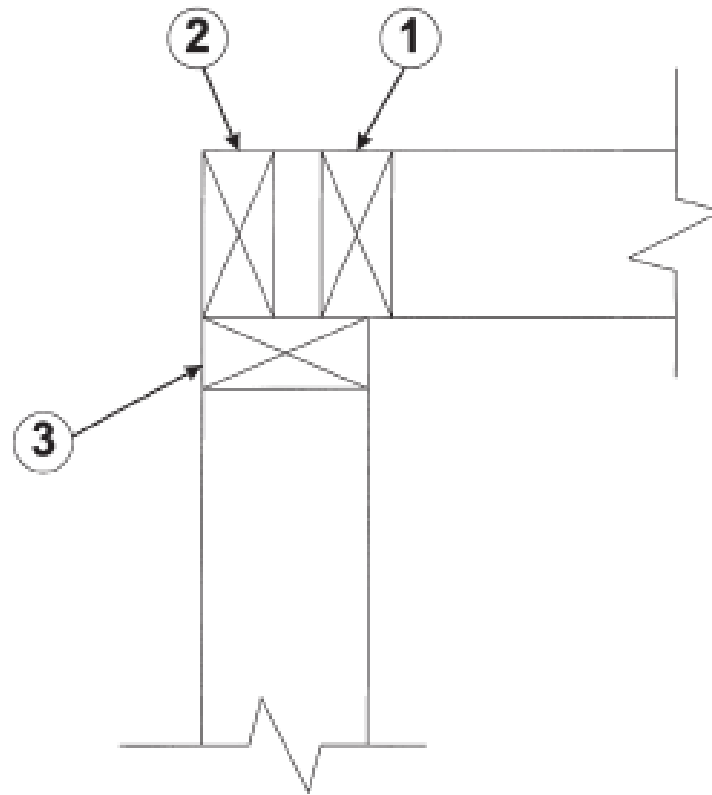
Studs

- Estimating
 - Review specification
 - Divide length by spacing
 - Add extra studs for:
 - Corners
 - Wall intersections
 - Wall openings
 - Add studs for gable ends

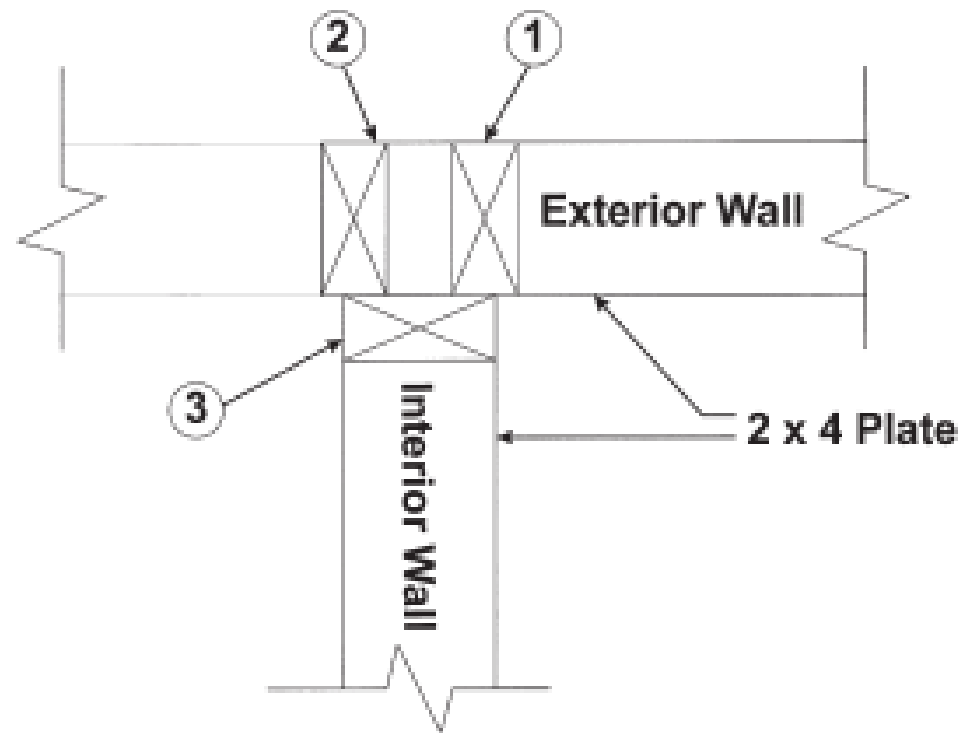
Studs

O.C. Spacing (inches)	Divide length to be framed by	Multiply length to be framed by
12	1	1
16	1.33	0.75
20	1.67	0.60
24	2	0.50
30	2.5	0.40
36	3	0.33

Corners



Wall Intersections



Determine the purchase quantity of 2 × 4 **studs** needed for the exterior wall. The studs are 8' long and placed at 16" on center. Add 2 studs per corner and intersection point with interior walls. Add 3 studs per opening.

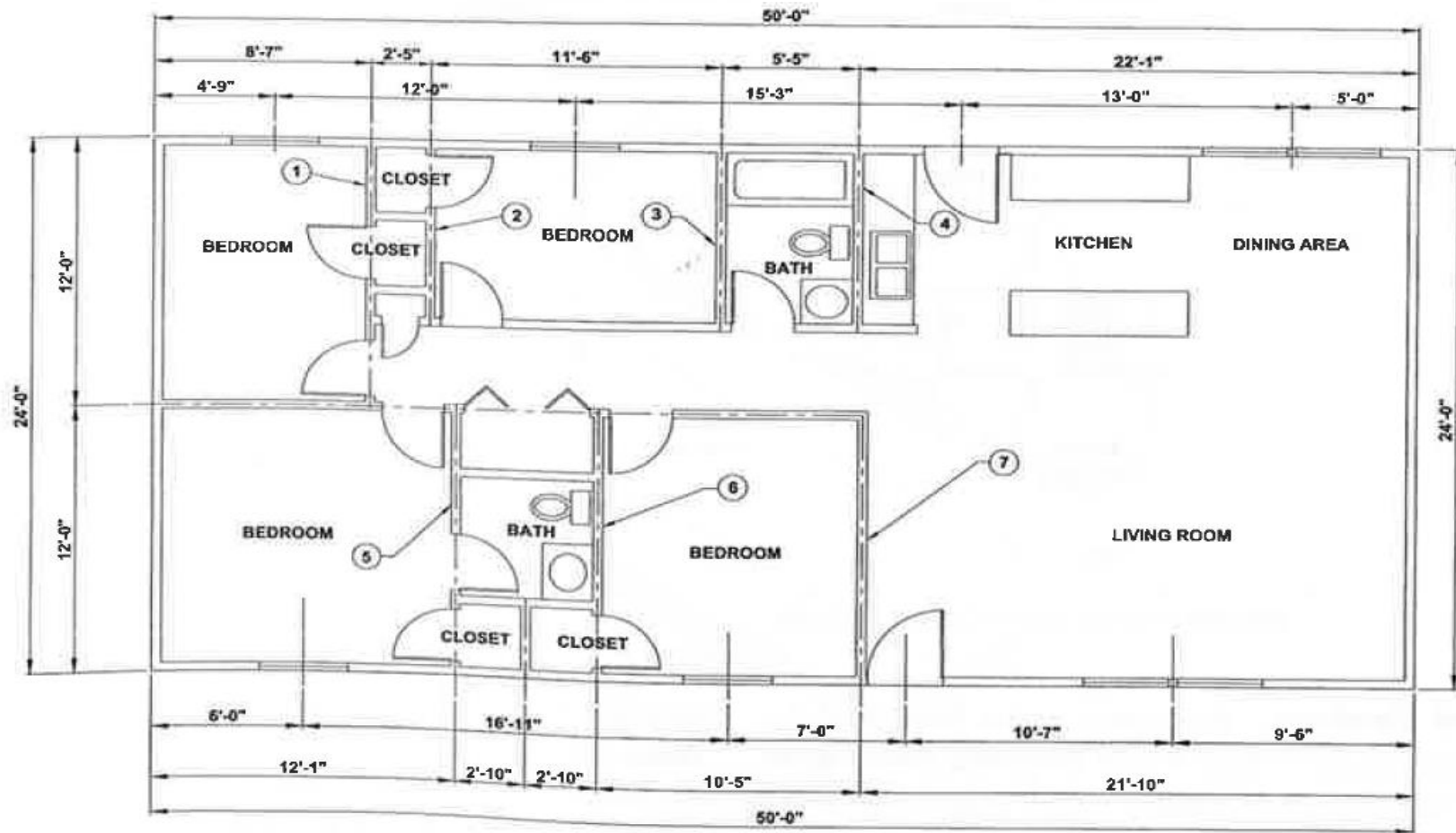


FIGURE 13.9. Floor Plan.

Exterior wall studs

= $148' \times 12/16 = 111$ spaces → Use 112 studs

Corner studs

= $4 \text{ corners} \times 2 = 8$ studs

Intersection with interior walls

= $9 \text{ intersections} \times 2 = 18$ studs

Openings

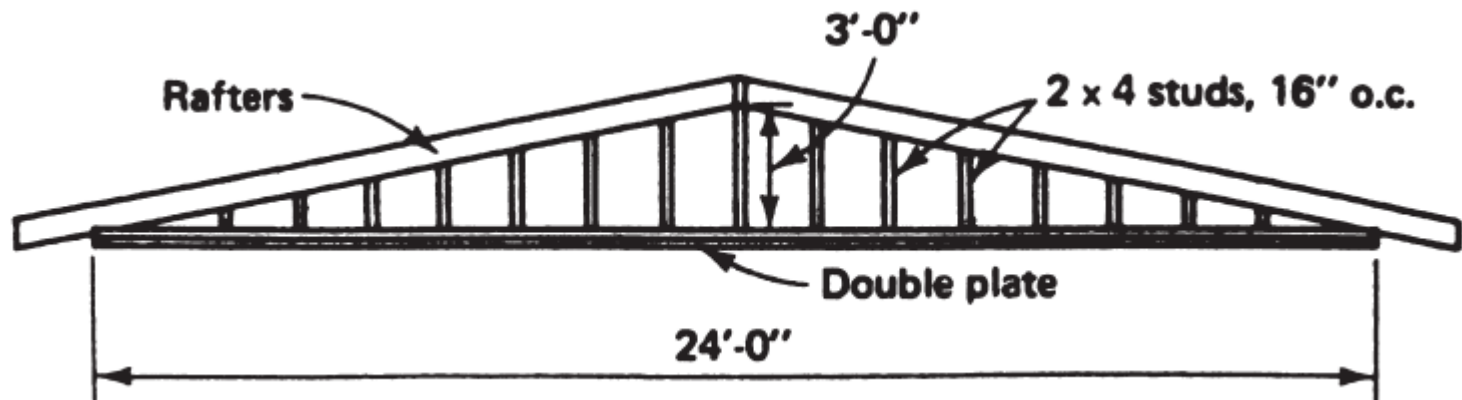
= $8 \text{ openings} \times 3 = 24$ studs

Purchase Quantity

$$= 112 + 8 + 18 + 24 = 162 \text{ studs}$$

162 pieces 2 × 4 @ 8'

Gable Ends



Determine the purchase quantity of 2 × 4 **studs** needed for the gable ends.

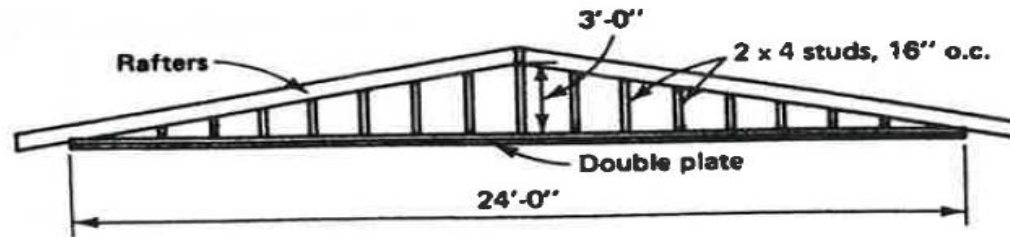
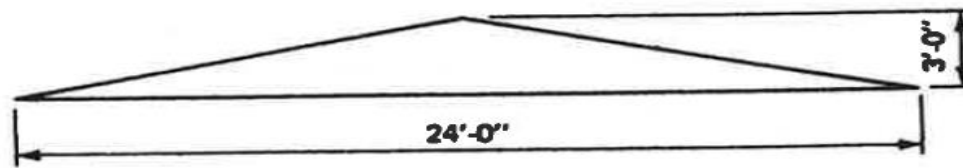
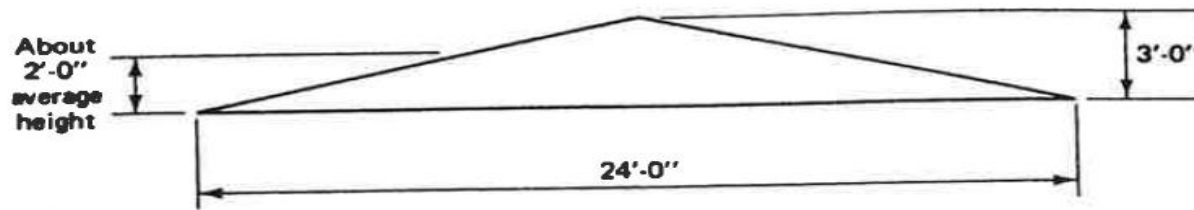


FIGURE 13.34. Gable End Frame.



Gable end

FIGURE 13.35. Gable End Sketch.



Gable end

FIGURE 13.36. Gable Average Height.

Gable end studs

= $24' \times 12/16 = 18$ spaces → Use 19 studs

Average height

= $2/3 \times h = 2/3 \times 3' = 2'$

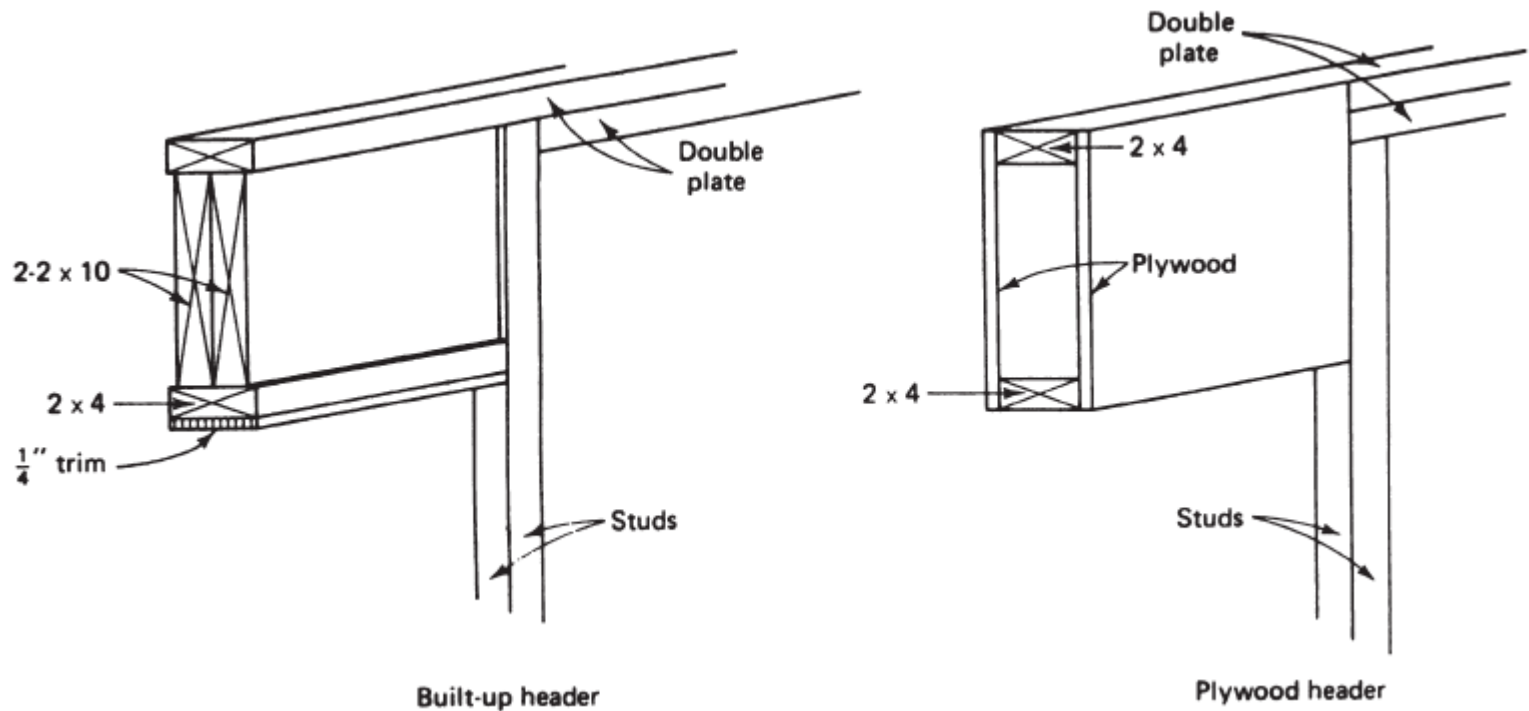
Quantity must be doubled (2 sides) = 38 studs

Purchase Quantity

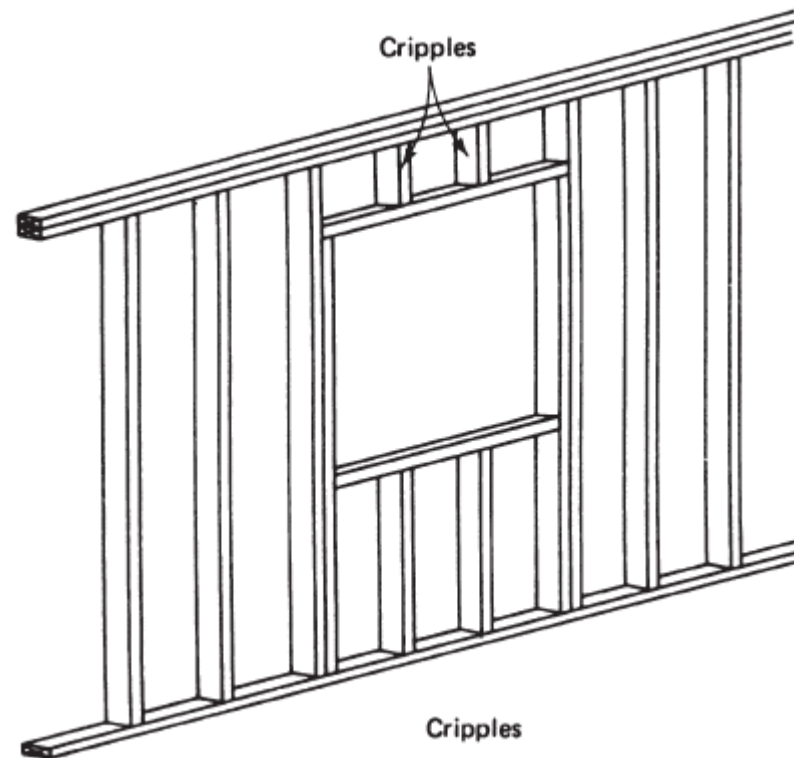
10 pieces 2×4 @ 8' or

8 pieces 2×4 @ 10' etc...

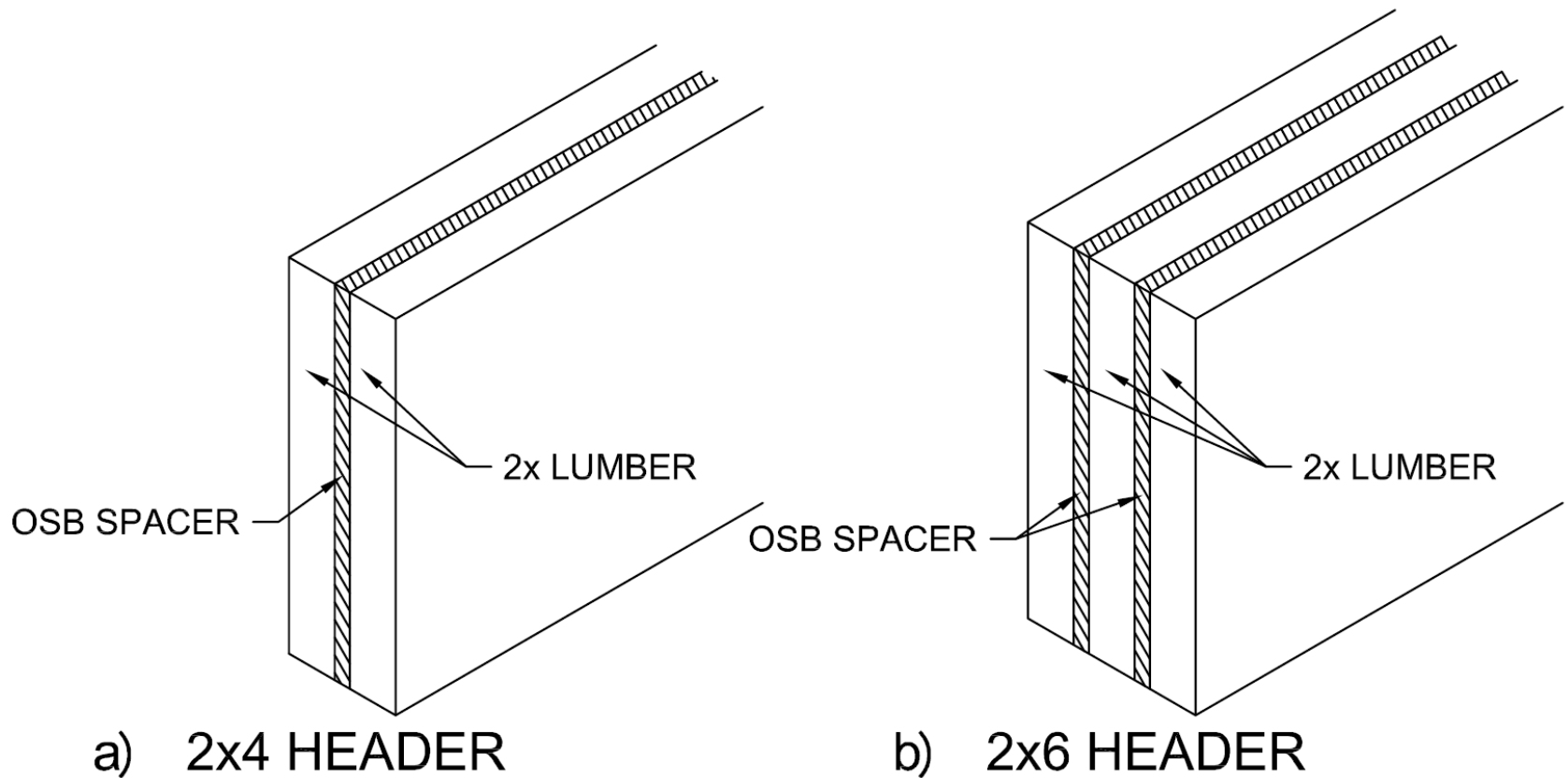
Headers



Headers



Headers



Exterior Wall Sheathing

- May be:
 - Fiberboard, bituminous impregnated
 - Insulation board
 - Waferboard (OSB)
 - Plywood

- Bid same as subflooring (sketch wall)

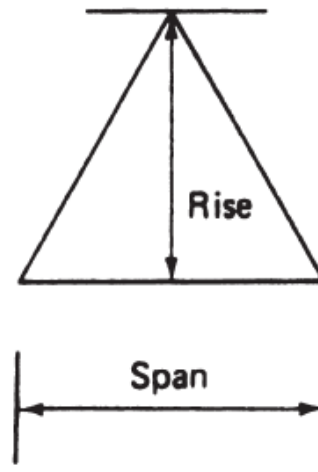
Ceiling Assembly

- Includes:
 - Ceiling joists
 - Headers and trimmers
- Bid same as floor joists
- Not needed with trusses

Roof Assembly

- Includes:
 - Rafters
 - Headers and trimmers
 - Collar ties
 - Ridge
 - Lookouts
 - Sheathing
 - Felt (may be included with roofing)

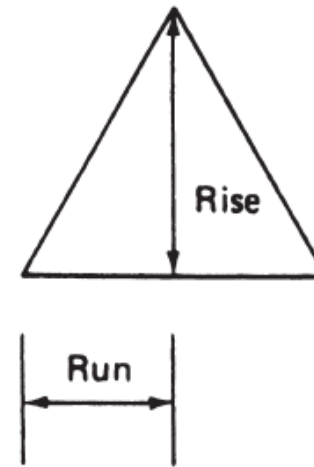
Slope and Pitch



Pitch

$$\text{Pitch} = \frac{\text{Rise}}{\text{Span}}$$

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}}$$



Slope

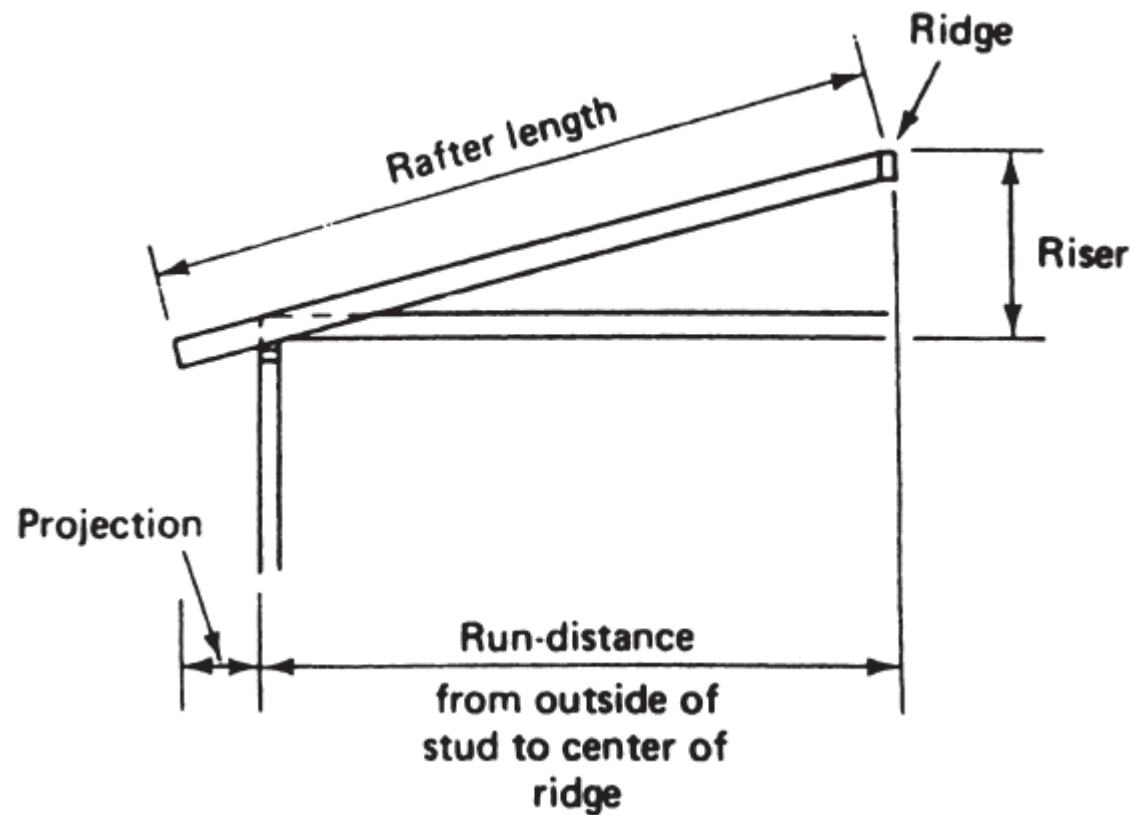
Rafters

- Must take into account the slope of the roof
- Estimating
 - Divide length by spacing
 - Handle openings the same as openings in floor or ceiling joists

Rafters

O.C. Spacing (inches)	Divide length to be framed by	Multiply length to be framed by
12	1	1
16	1.33	0.75
20	1.67	0.60
24	2	0.50
30	2.5	0.40
36	3	0.33

Rafter Length



Rafter Length

Pitch of Roof	Slope of Roof	For length of rafter multiply length of run by
1/12	2 in 12	1.015
1/8	3 in 12	1.03
1/6	4 in 12	1.055
5/24	5 in 12	1.083
1/4	6 in 12	1.12

Note: Run measurement should include any required overhang

Determine the purchase quantity of 2 × 8 **rafters** needed for the roof. The slope is 3 on 12, projection length is 1' 6", and 1' overhang to each side.

$$\text{Run} = 12' + 1' 6'' = 13.5'$$

$$\text{Rafter length} = 13.5' \times 1.03 = 13.9' \rightarrow 14'$$

$$\text{Ridge length} = 50' + 2 \times 1' = 52'$$

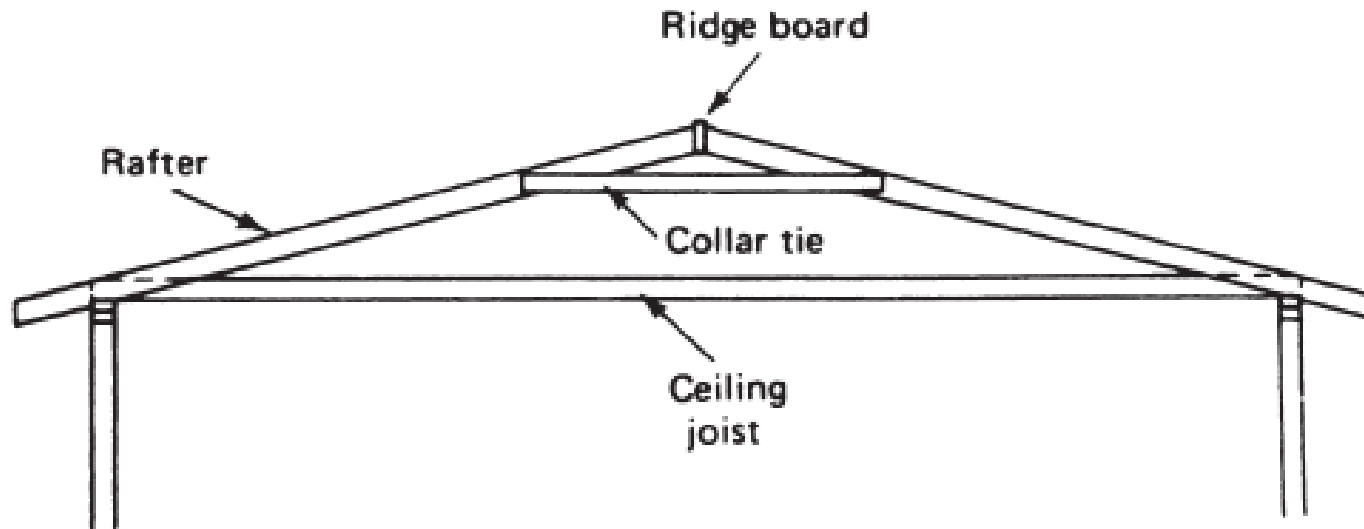
$$\begin{aligned} &\text{Rafter quantity per side} \\ &= 52' \times 12/16 + 1 = 40 \end{aligned}$$

There are 2 sides, so 80 rafters are needed.

Purchase Quantity

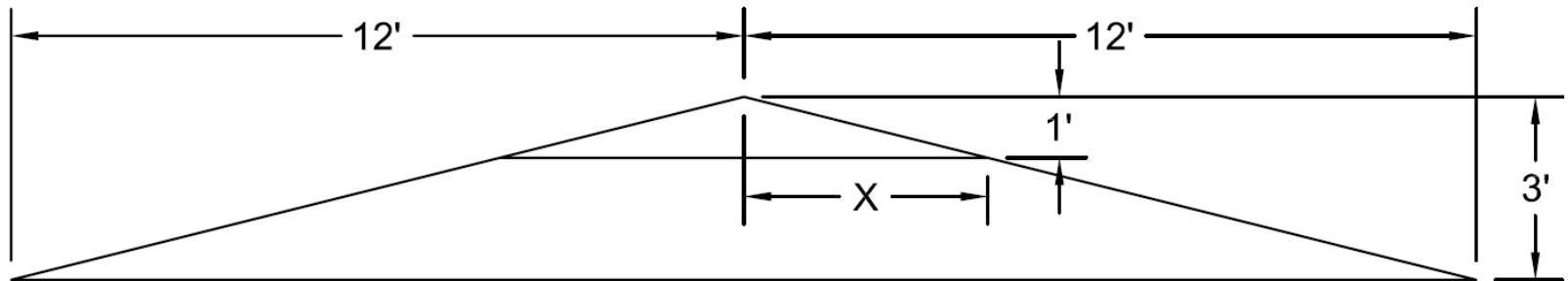
80 pieces 2 × 8 @ 14'

Collar Ties and Ridge Board



Collar Ties

- Usually lesser of every third rafter or 5 feet



SIMILAR TRINAGLES

$$3/12 = 1/X$$

$$X = 4'$$

$$\text{Collar Length} = 8'$$

Determine the purchase quantity of 2 × 4 **collar ties** needed for the roof.

Spaced at the maximum of 5' or every third rafter, whichever is less. So, install at the every third rafters. → 14 collar ties

Collar tie length → $24' : 3' = x : 1'$
→ $x = 8'$

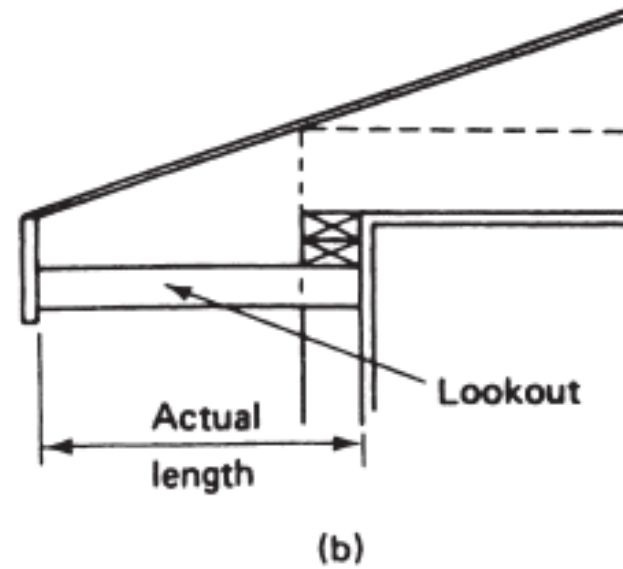
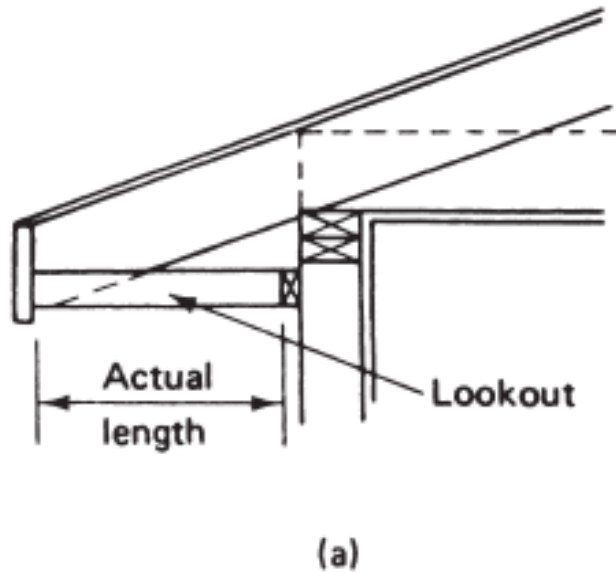
Purchase Quantity

14 pieces 2 × 4 @ **8'**

Ridge

- Equals the length of the building plus any overhang

Lookouts



Roof Sheathing

- Be sure to take slope into account
- Sketch sheathing as was done for subflooring and wall sheathing

Trim

- Exterior:

- Moldings
- Fascias
- Cornices
- Corner boards

- Interior

- Base
- Moldings
- Chair rails
- Other

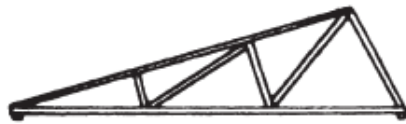
Framing Labor Productivity

Light Framing	Unit	Labor Hours
Sills	100 lf	2.0 to 4.5
Joists, floors and ceilings	MBM	16.0 to 24.0
Walls, interior, exterior (including plates)	MBM	18.0 to 30.0
Rafters, gable roof	MBM	18.5 to 30.0
Rafters, hip roof	MBM	22.0 to 35.0
Cross bridging, wood	100 sets	4.0 to 6.0
Cross bridging, metal	100 sets	3.0 to 5.0

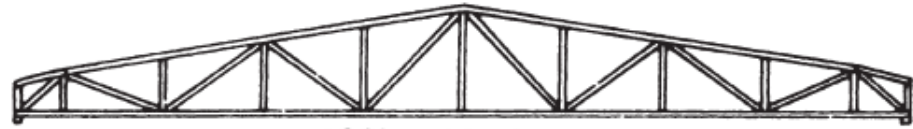
Framing Labor Productivity

Light Framing	Unit	Labor Hours
Plywood, floor	100 sf	1.0 to 2.0
Plywood, wall	100 sf	1.2 to 2.5
Plywood, roof	100 sf	1.4 to 2.8
Trim, fascia	100 lf	3.5 to 5.0
Trim, soffit	100 lf	2.0 to 3.5
Trim, baseboard	100 lf	1.5 to 2.5
Trim, molding	100 lf	2.0 to 4.0

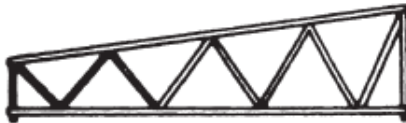
Wood Trusses



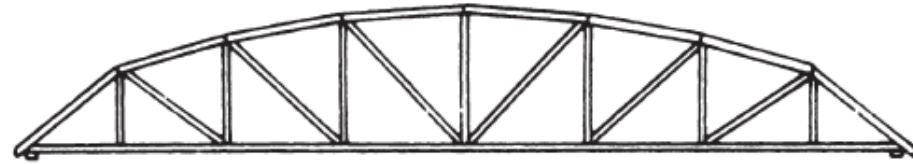
Dual pitch



100' stubbed truss



Skillion warren truss



Bowstring truss



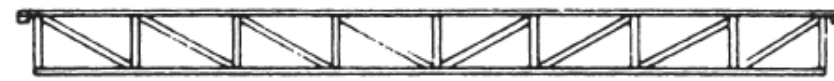
Hip louvre truss



Cantilevered flat roof truss

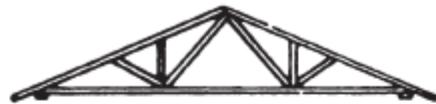


Modified queenpost truss

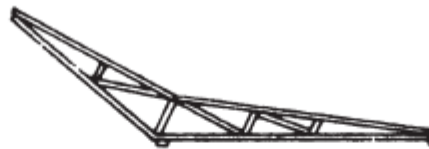


Flat floor truss

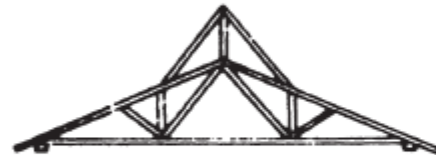
Wood Trusses



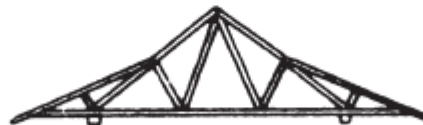
Fan truss



Cantilever shelter



Piggyback truss



Inverted scissors – field spliced

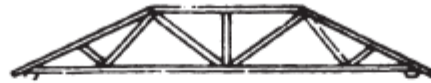


Air conditioner truss

Wood Trusses



Double fink truss



Hip truss



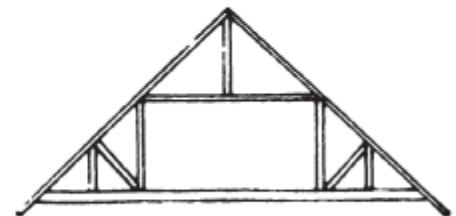
Scissors truss



Standard fink truss



Trussed frame



Attic truss



Inverted truss

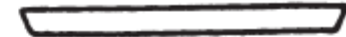
Laminated Beams and Arches



Symmetrically tapered beams



Curved tapered beams

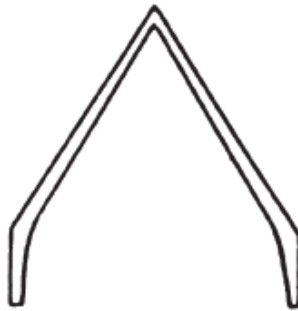


Straight beams



Special beam sections

Laminated Beams and Arches



High V



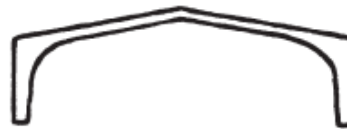
Low V



A



Parabolic arch



Hinged arch



Bowstring arch